CRYPTOPULSE AI

Complete Beginner Deployment and Operations Manual

Oracle Cloud + GitHub + Gemini + CoinGecko + Meta Publishing + AI Inbox

Manual v2.0 | Application v0.2.0 | Prepared 20 July 2026 | Beginner edition

Safety rule - Keep DRY_RUN=true until every test in this manual passes. Never paste a real API key into GitHub, screenshots, chat messages, or source code.

How to use this manual

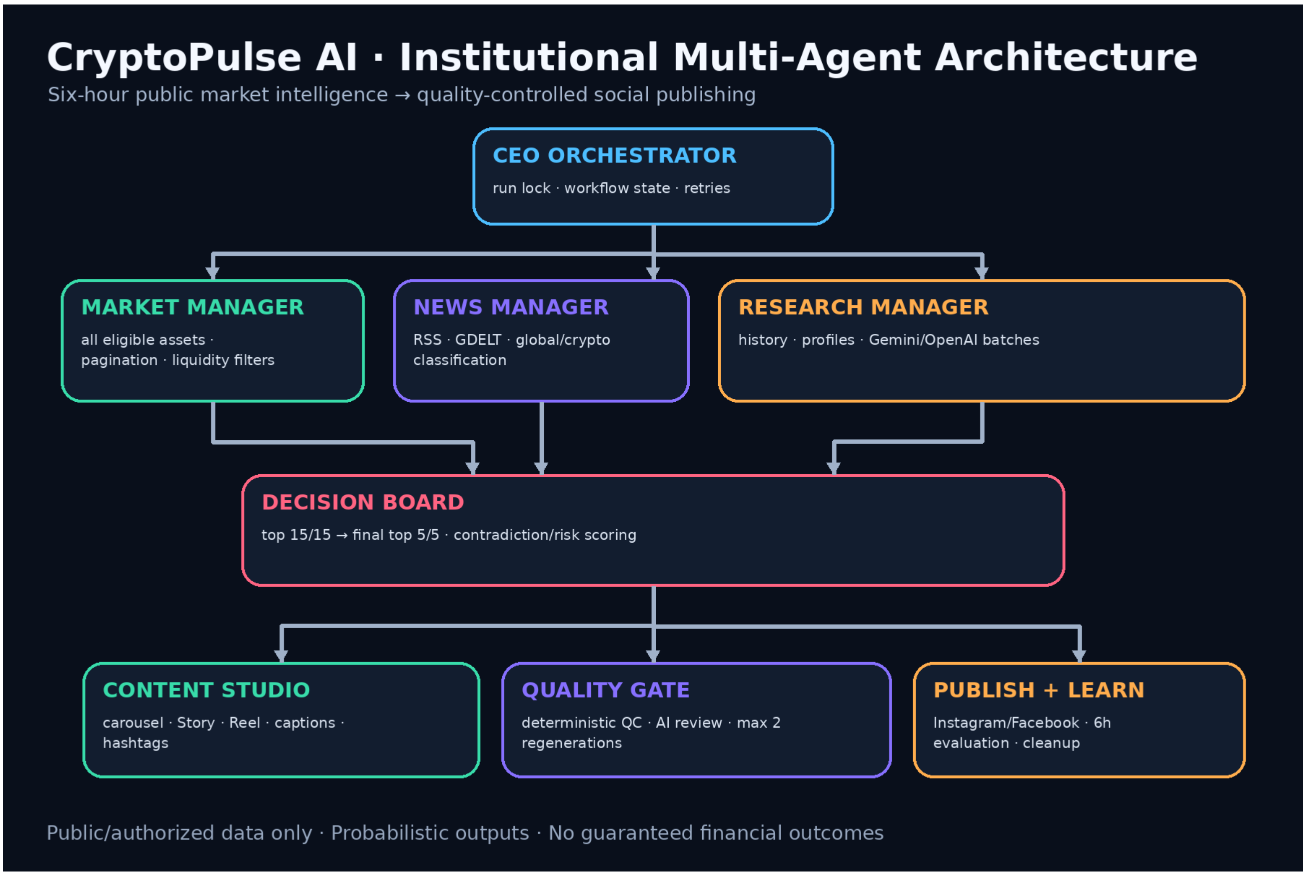
Follow the chapters in order. Every command is marked by the computer on which it must be run. Copy one command at a time, replace only the placeholders written in CAPITAL LETTERS, and confirm the expected result before continuing.

Label	Where you run it	Example
WINDOWS POWERSHELL	Your Windows PC	PS> git --version
ORACLE TERMINAL	The Ubuntu server after SSH login	ubuntu$ docker compose ps
BROWSER	Chrome/Edge on PC or phone	Oracle, GitHub, Google AI Studio, Meta
.ENV FILE	/opt/cryptopulse/.env on Oracle	GEMINI_API_KEY=...

Placeholders - Do not type angle brackets. Replace YOUR_USERNAME, YOUR_PUBLIC_IP, YOUR_GEMINI_KEY, and similar labels with your own values.

Table of contents

1. Understand the system and safety gates



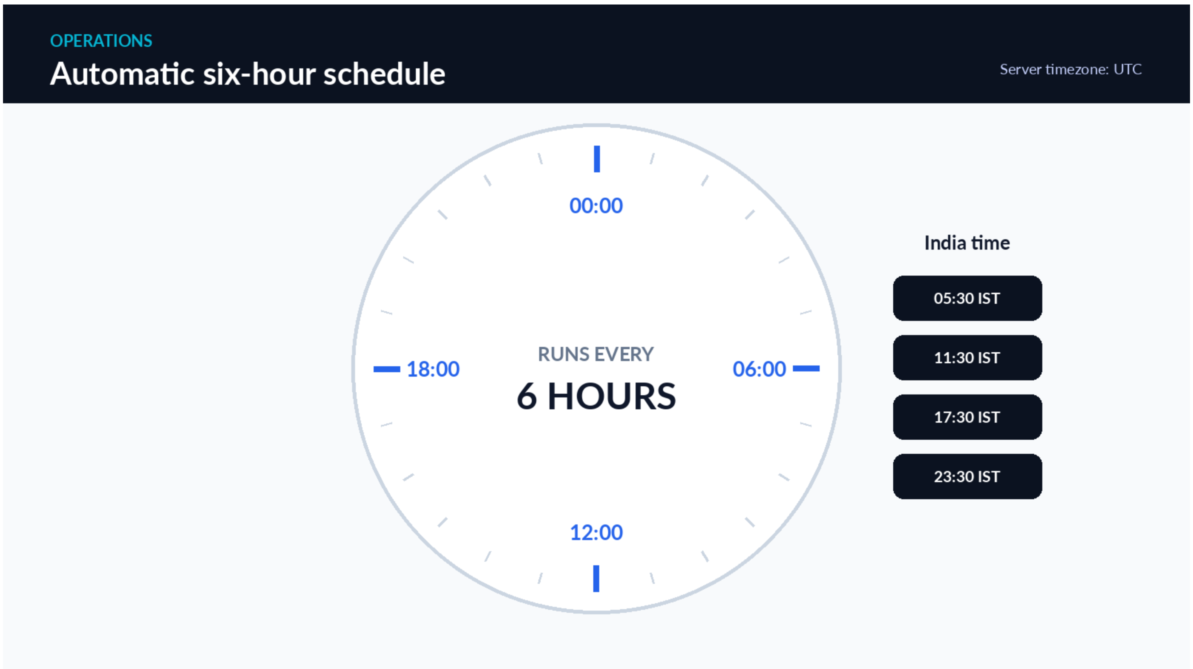
The application uses logical CEO, manager and worker agents inside a controlled workflow.

The Oracle server does not run large AI models locally. It coordinates market-data calls, news collection, Gemini analysis, ranking, image and video generation, quality checks, publishing, evaluation, and cleanup. This is why a 2 OCPU / 12 GB A1 Flex server is suitable for the zero-budget starting architecture.

1.1 What happens every six hours

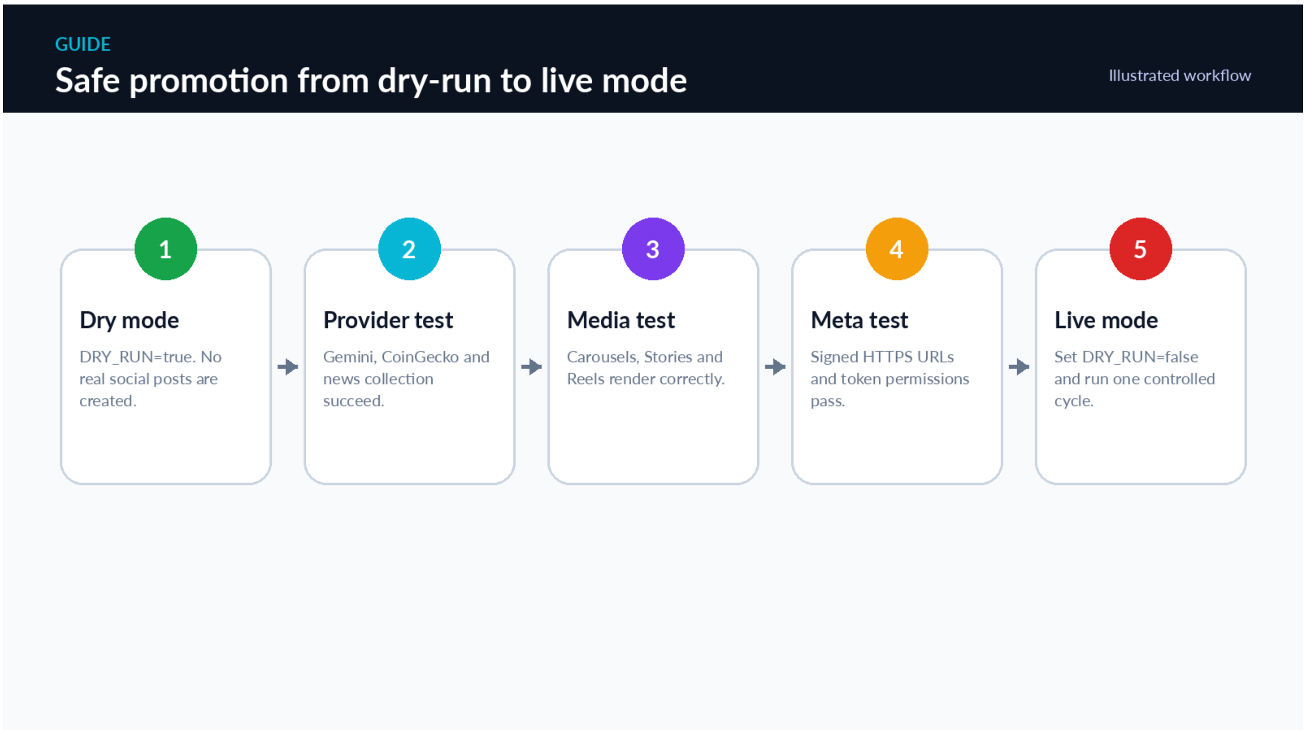
1	The scheduler starts one run at 00:00, 06:00, 12:00 and 18:00 UTC.
2	Market and news workers collect structured data simultaneously.
3	The system scores the market and shortlists 15 bullish and 15 bearish candidates.
4	Deep research and the final decision board select five Bullish Watch and five Bearish Risk assets.
5	The content studio creates two carousels, Stories, optional Reels, captions and hashtags.
6	Quality control verifies facts, layout, wording and publishing requirements.
7	In live mode, approved content is posted to Instagram and Facebook.

8	Temporary files are deleted after the configured grace period; minimal prediction records are retained.



1.2 Four operating modes you must understand

Mode	Important settings	What it does
Offline sample	heuristic + sample command	Creates local sample media without live APIs.
Dry run	DRY_RUN=true	Uses configured providers but does not publish real posts.
Controlled live	DRY_RUN=false + run-once	Publishes one manually started cycle.
Scheduled live	DRY_RUN=false + service running	Publishes automatically every six hours.



Promotion rule - Never jump directly from installation to scheduled live mode. Complete offline sample, dry-run, public HTTPS, Meta token and controlled live gates first.

2. Prepare Windows and the release package

CHECKLIST

What you need before starting

No programming experience is required

- **Windows PC** — Windows 10 or 11 with administrator access.
- **Internet** — Stable connection for downloads and SSH.
- **GitHub account** — Stores the source code safely.
- **Oracle account** — Hosts the production server.
- **Google account** — Creates the Gemini API key.
- **Meta assets** — Instagram professional account and Facebook Page.
- **Payment card** — Oracle may request it only for identity verification.
- **Two hours** — Do not rush the first deployment.

2.1 Create one project folder

1	Download cryptopulse_ai_oracle_release_v0.1.0.zip from this ChatGPT conversation.
2	Open File Explorer and go to Downloads.
3	Right-click the ZIP file, choose Extract All, and extract it into a folder named cryptopulse_release.
4	Open the extracted cryptopulse_ai folder and confirm that README.md, Dockerfile, docker-compose.yml, src, scripts and docs are present.

2.2 Install Git on Windows

WINDOWS POWERSHELL - run as your normal Windows user

```
winget install --id Git.Git -e --source winget
```

Close PowerShell, open it again, and verify Git:

```
git --version
```

Expected result: a line similar to git version 2.x.x. The exact version can be newer.

2.3 Configure your Git name and email

```
git config --global user.name "Sunny"
git config --global user.email "YOUR_GITHUB_EMAIL"
```

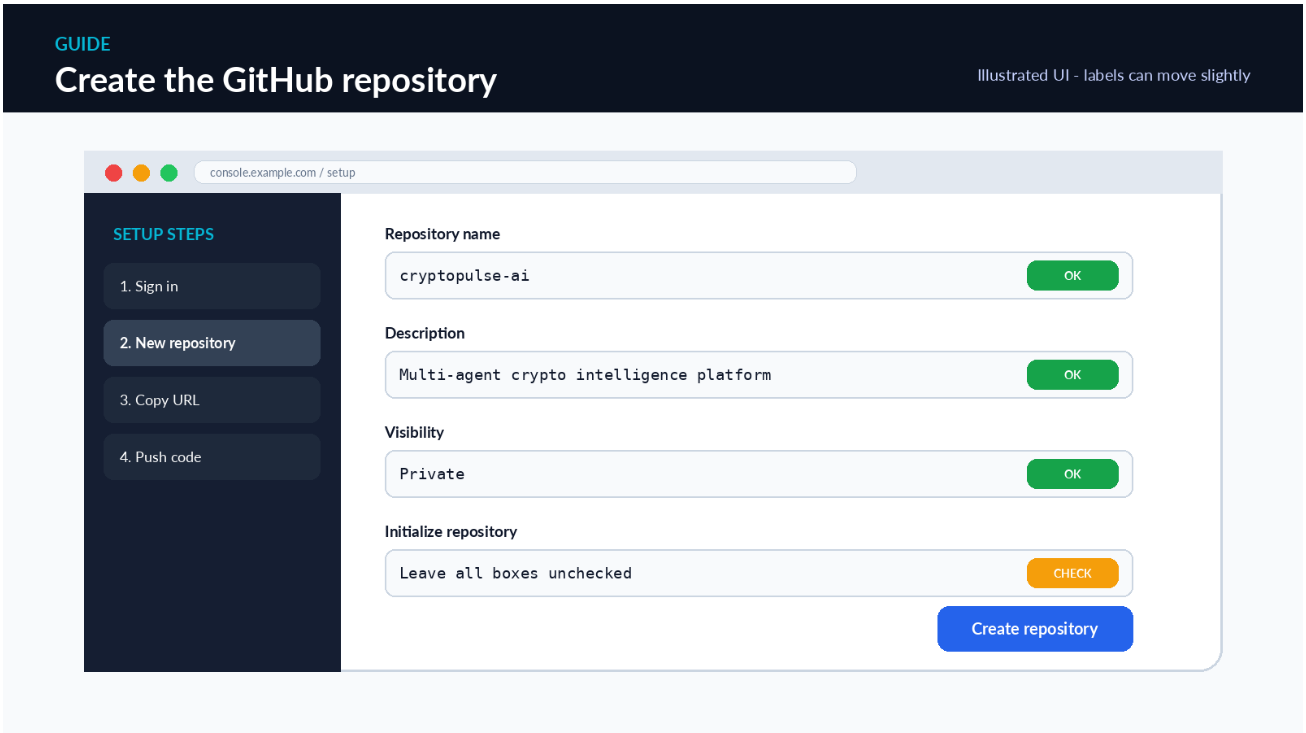
> **Privacy option -** GitHub can provide a no-reply email address. You may use that address instead of your personal email.

2.4 Verify that secrets are not present

The release contains .env.example, not a real .env file. Open .gitignore and confirm that .env is listed. Real keys will be created only on Oracle.

```
cd "$HOME\Downloads\cryptopulse_release\cryptopulse_ai"
Get-ChildItem -Force
Get-Content .gitignore
```

3. Create GitHub and upload the code



Illustrated GitHub screen. The live interface may move buttons, but the field names remain recognizable.

3.1 Create the repository in your browser

1	Sign in to GitHub and click the plus (+) menu in the upper-right corner.
2	Choose New repository.
3	Enter repository name cryptopulse-ai.
4	Select Private.
5	Do not add a README, .gitignore or license because the release already contains them.
6	Click Create repository and keep the Quick setup page open.

Why private? - A private repository protects your application code. The .env file is still kept out of GitHub, even in a private repository.

3.2 Push the extracted project from PowerShell

TERMINAL

Upload the release code from Windows PowerShell

Copy one command at a time

PowerShell / OCI Ubuntu terminal

```
PS>    cd "$HOME\Downloads\cryptopulse_release\cryptopulse_ai"
PS>    git init && git add .
PS>    git commit -m "Initial CryptoPulse AI release"
PS>    git branch -M main
PS>    git remote add origin https://github.com/YOUR_USERNAME/cryptopulse-ai.git
PS>    git push -u origin main
```

```
cd "$HOME\Downloads\cryptopulse_release\cryptopulse_ai"
git init
git add .
git commit -m "Initial CryptoPulse AI release"
git branch -M main
git remote add origin https://github.com/YOUR_USERNAME/cryptopulse-ai.git
git remote -v
git push -u origin main
```

During the first push, Git Credential Manager may open a browser window. Sign in to GitHub and authorize Git. GitHub account passwords are not used directly for Git pushes.

3.3 Confirm the upload

1	Refresh your repository page in the browser.
2	Confirm that folders such as src, scripts, infra, docs and tests are visible.
3	Confirm that .env is not present.
4	Open Actions and confirm that the CI workflow starts. A green check is the expected final result.

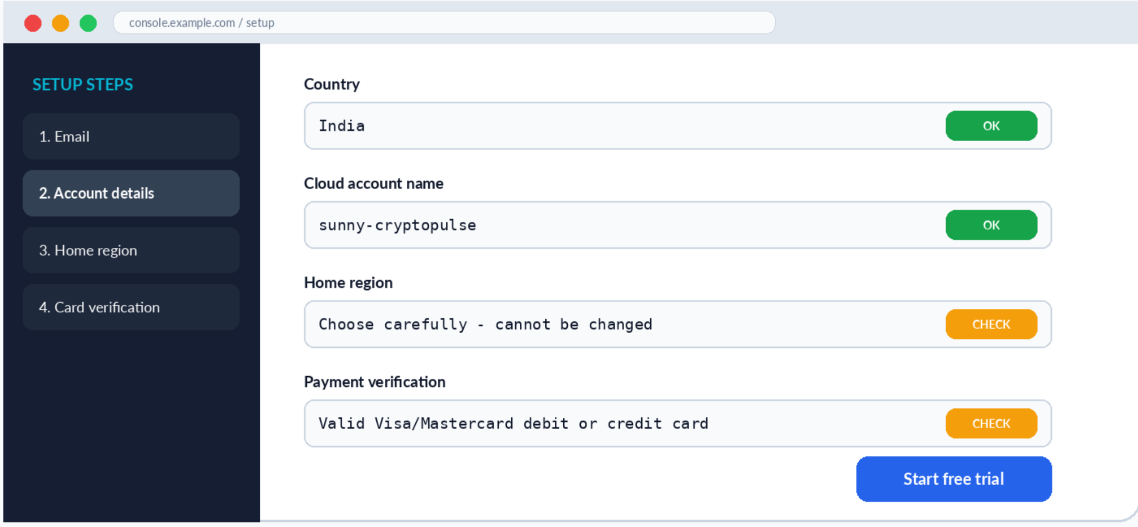
Push rejected? - If the remote repository was initialized with a README, create a new empty repository or carefully pull and merge. The beginner-safe method is to create the repository without initialization.

4. Create the Oracle Cloud account

GUIDE

Create the Oracle Cloud account

Illustrated UI - labels can move slightly



Oracle currently documents a free-tier sign-up flow with email verification, a permanent home-region choice and payment-card identity verification. A small temporary authorization hold may appear; Oracle states that you are not charged unless you upgrade the account. [S1]

4.1 Sign up

1	Open the Oracle Cloud Free Tier sign-up page.
2	Choose India as your country and enter your real name and email.
3	Complete the human verification and click Verify my email.
4	Open the Oracle email within 30 minutes and follow the verification link.
5	Create a strong password and a memorable cloud account name.
6	Choose the home region carefully. Oracle states that it cannot be changed after sign-up. [S1]
7	Enter your address and phone number.
8	Add a supported credit or debit card for identity verification and complete the sign-up.

Home region - Choose an Indian region available in the sign-up flow. Capacity for A1 Flex may vary. A home region cannot later be changed, so do not choose randomly.

4.2 Record these values

Value	Your entry
Cloud account name	____
Home region	____
Oracle login email	____
Date created	____

Oracle signup

https://signup.oraclecloud.com

Scan to open the official Oracle sign-up page.

5. Create the Oracle network and server

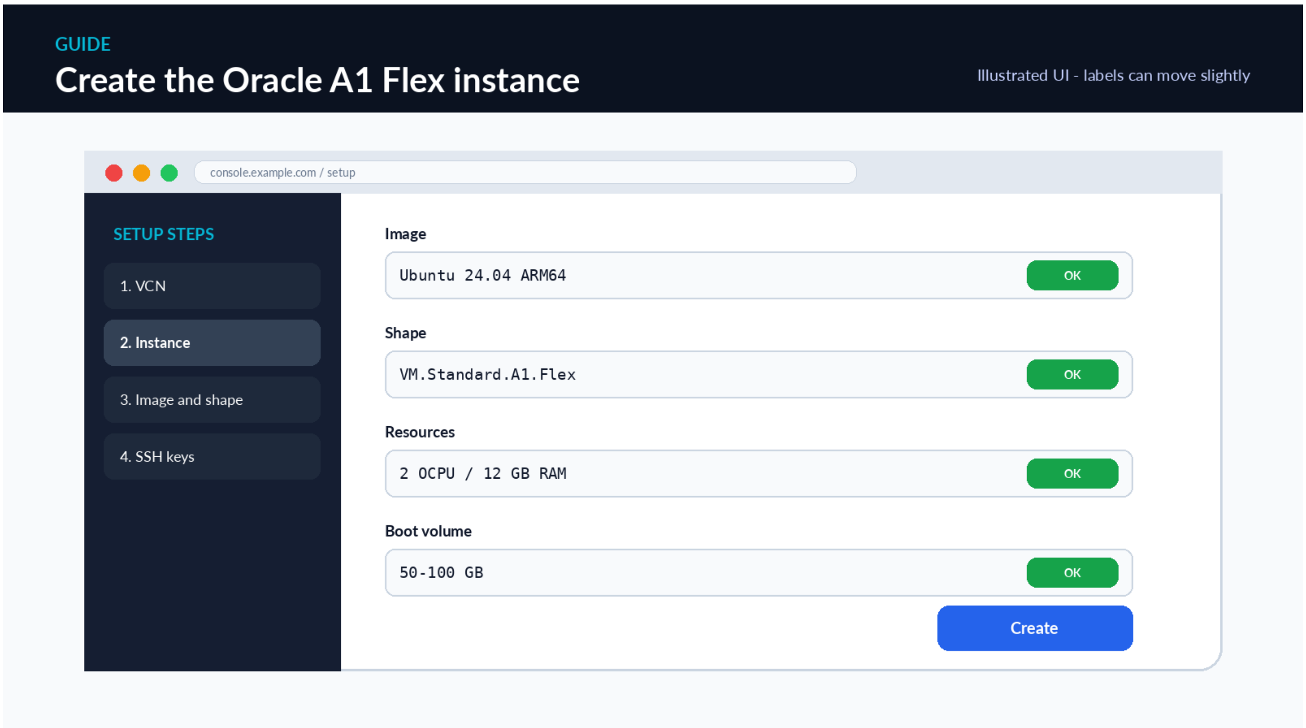
Oracle documents the Always Free A1 allocation as the equivalent of 2 OCPUs and 12 GB of memory in total, subject to capacity and tenancy rules. The minimum boot volume is 47 GB. [S2]

5.1 Create a VCN with internet connectivity

1	Open the Oracle navigation menu.
2	Choose Networking, then Virtual Cloud Networks.
3	Start the VCN Wizard and choose Create VCN with Internet Connectivity.
4	Name it cryptopulse-vcn and keep the generated public subnet, internet gateway, route table and DNS settings.
5	Click Next, review and create the VCN.

> **Why a public subnet? -** The application must receive HTTPS requests from Meta so Instagram and Facebook can download generated media.

5.2 Create the compute instance



1	Go to Compute, then Instances, and click Create instance.
2	Name the instance cryptopulse-prod.

3	Choose the cryptopulse-vcn public subnet and enable a public IPv4 address.
4	Change image to Ubuntu 24.04 ARM64 or the latest supported Ubuntu ARM64 image.
5	Change shape to Ampere, VM.Standard.A1.Flex.
6	Set 2 OCPUs and 12 GB RAM.
7	Set the boot volume to 50-100 GB.
8	Choose Generate a key pair for me and download both private and public key files.
9	Click Create and wait for the instance state to become Running.

Protect the private key - The .key file is the login credential for your server. Keep at least one encrypted backup. Do not upload it to Google Drive shared folders, GitHub or social media.

5.3 Record the instance details

Value	Your entry
Instance name	cryptopulse-prod
Public IPv4 address	______________________________________
Shape	VM.Standard.A1.Flex
OCPU / memory	2 / 12 GB
Private key path	C:\Users\YOUR_NAME\.ssh\cryptopulse_oci.key

5.4 Add ingress rules

GUIDE

Open only the required network ports

Illustrated UI - labels can move slightly

1	Open the instance, click its subnet, then open the security list used by that subnet.
2	Add TCP port 22. Prefer your current public IP followed by /32. If your internet IP changes frequently, temporarily use 0.0.0.0/0 and rely on SSH key authentication, then restrict it later.
3	Add TCP port 80 from 0.0.0.0/0.
4	Add TCP port 443 from 0.0.0.0/0.
5	Do not add public access to 8080, SQLite or any database port.

6. Connect to Oracle from Windows

Oracle supports connecting to Ubuntu instances from Windows PowerShell with OpenSSH. The username for Ubuntu images is ubuntu. The private key file must be restricted to your Windows account. [S3]

6.1 Move and secure the private key

TERMINAL

Connect from Windows PowerShell

Copy one command at a time

PowerShell / OCI Ubuntu terminal

```
PS>   mkdir "$HOME\.ssh" -ErrorAction SilentlyContinue
PS>   Move-Item "$HOME\Downloads\ssh-key-*.key" "$HOME\.ssh\cryptopulse_oci.key"
PS>   icacls "$HOME\.ssh\cryptopulse_oci.key" /inheritance:r
PS>   icacls "$HOME\.ssh\cryptopulse_oci.key" /grant:r "$($env:USERNAME):(F)"
PS>   ssh -i "$HOME\.ssh\cryptopulse_oci.key" ubuntu@YOUR_PUBLIC_IP
```

WINDOWS POWERSHELL

```
New-Item -ItemType Directory -Force "$HOME\.ssh"
Move-Item "$HOME\Downloads\YOUR_DOWNLOADED_KEY.key" "$HOME\.ssh\cryptopulse_oci.key"
icacls "$HOME\.ssh\cryptopulse_oci.key" /inheritance:r
icacls "$HOME\.ssh\cryptopulse_oci.key" /grant:r "$($env:USERNAME):(F)"
```

Replace YOUR_DOWNLOADED_KEY.key with the exact downloaded filename. If Move-Item says the file does not exist, open Downloads and copy its exact name.

6.2 Connect

```
ssh -i "$HOME\.ssh\cryptopulse_oci.key" ubuntu@YOUR_PUBLIC_IP
```

On the first connection, type yes when asked whether you trust the host fingerprint. A successful login changes the prompt to something similar to ubuntu@cryptopulse-prod:~$.

6.3 Verify the server

```
uname -m
cat /etc/os-release
free -h
df -h
```

Command	Expected
uname -m	aarch64 or arm64
cat /etc/os-release	Ubuntu information
free -h	Approximately 12 GB total memory
df -h	At least about 47 GB root volume

Permission denied (publickey) - Usually means the wrong private key, wrong username, key file permissions, or port 22 security rule. Do not create a password login as a shortcut.

7. Connect Oracle to the private GitHub repository

A GitHub deploy key gives this Oracle server read-only access to one private repository. It is safer than storing your personal GitHub password or personal access token on the server.

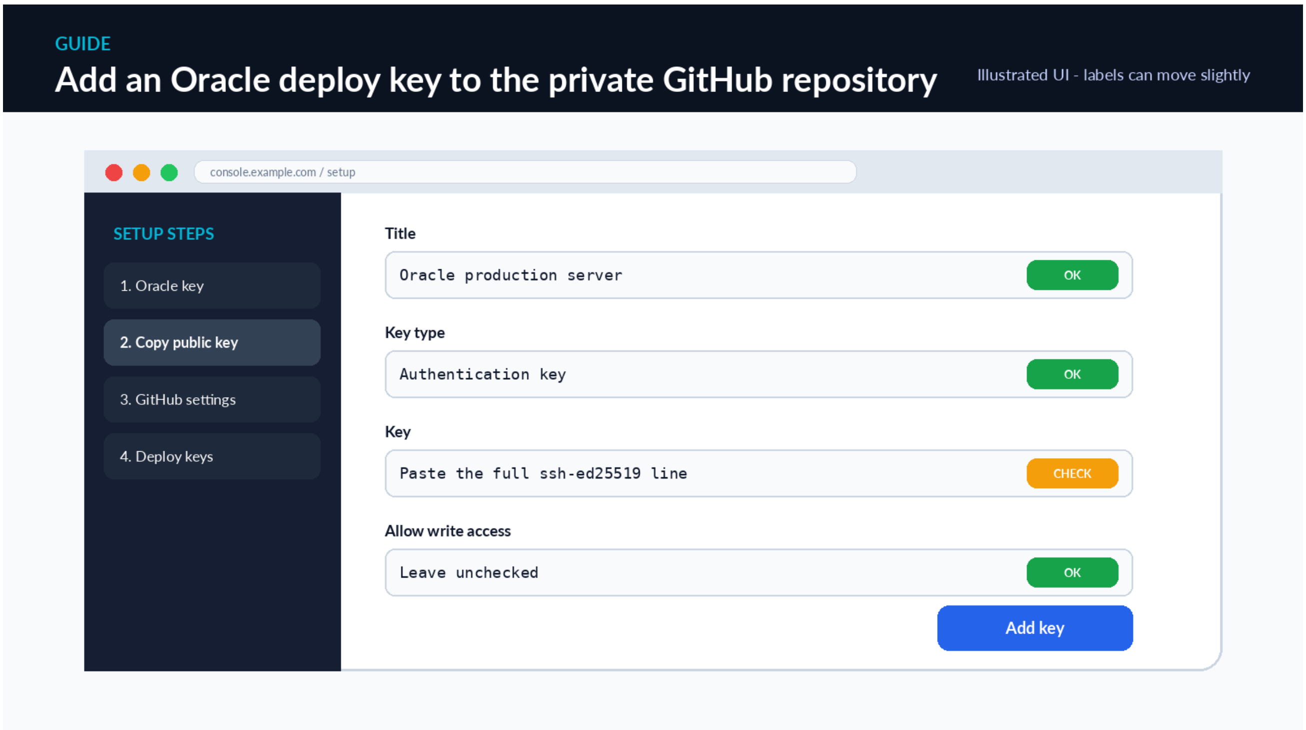
7.1 Generate the deploy key on Oracle

ORACLE TERMINAL

```
mkdir -p ~/.ssh
chmod 700 ~/.ssh
ssh-keygen -t ed25519 -C "cryptopulse-oracle" -f ~/.ssh/cryptopulse_deploy -N ""
cat ~/.ssh/cryptopulse_deploy.pub
```

Copy the entire output line beginning with ssh-ed25519. Copy only the public .pub key. Never copy the private file named cryptopulse_deploy.

7.2 Add it to GitHub



1	Open your cryptopulse-ai repository in GitHub.
2	Click Settings, then Deploy keys.
3	Click Add deploy key.
4	Title: Oracle production server.
5	Paste the full ssh-ed25519 public key.
6	Leave Allow write access unchecked.

7	Click Add key.

7.3 Configure SSH on Oracle

```
cat > ~/.ssh/config <<'EOF'
Host github.com
  HostName github.com
  User git
  IdentityFile ~/.ssh/cryptopulse_deploy
  IdentitiesOnly yes
EOF
chmod 600 ~/.ssh/config
ssh-keyscan github.com >> ~/.ssh/known_hosts
chmod 644 ~/.ssh/known_hosts
ssh -T git@github.com
```

A successful test says that you authenticated successfully and that GitHub does not provide shell access. That message is normal.

> **Repository not found -** Check the deploy key is attached to the correct repository and the clone URL uses the exact GitHub username and repository name.

8. Bootstrap and deploy the application

TERMINAL

Bootstrap and deploy on Oracle

Copy one command at a time

PowerShell / OCI Ubuntu terminal

```
ubuntu$ git clone git@github.com:YOUR_USERNAME/cryptopulse-ai.git /tmp/cryptopulse-source
ubuntu$ sudo bash /tmp/cryptopulse-source/scripts/bootstrap_oci.sh
ubuntu$ exit
PS>     ssh -i "$HOME\.ssh\cryptopulse_oci.key" ubuntu@YOUR_PUBLIC_IP
ubuntu$ bash /tmp/cryptopulse-source/scripts/deploy_oci.sh git@github.com:YOUR_USERNAME/cryptopulse-
  ai.git main
```

8.1 Clone the source into /tmp

ORACLE TERMINAL

```
git clone git@github.com:YOUR_USERNAME/cryptopulse-ai.git /tmp/cryptopulse-source
```

8.2 Install Docker, firewall rules and the application directory

```
sudo bash /tmp/cryptopulse-source/scripts/bootstrap_oci.sh
```

This script installs Docker Engine, Docker Compose, Git, certificate tools and UFW. It allows SSH, HTTP and HTTPS; creates /opt/cryptopulse; and adds the ubuntu user to the docker group.

> **Required logout -** Group membership is refreshed only after you log out and reconnect. Do not skip the next two commands.

ORACLE TERMINAL

```
exit
```

WINDOWS POWERSHELL

```
ssh -i "$HOME\.ssh\cryptopulse_oci.key" ubuntu@YOUR_PUBLIC_IP
```

8.3 Deploy from Git

ORACLE TERMINAL

```
bash /tmp/cryptopulse-source/scripts/deploy_oci.sh git@github.com:YOUR_USERNAME/cryptopulse-
ai.git main
cd /opt/cryptopulse
docker compose ps
```

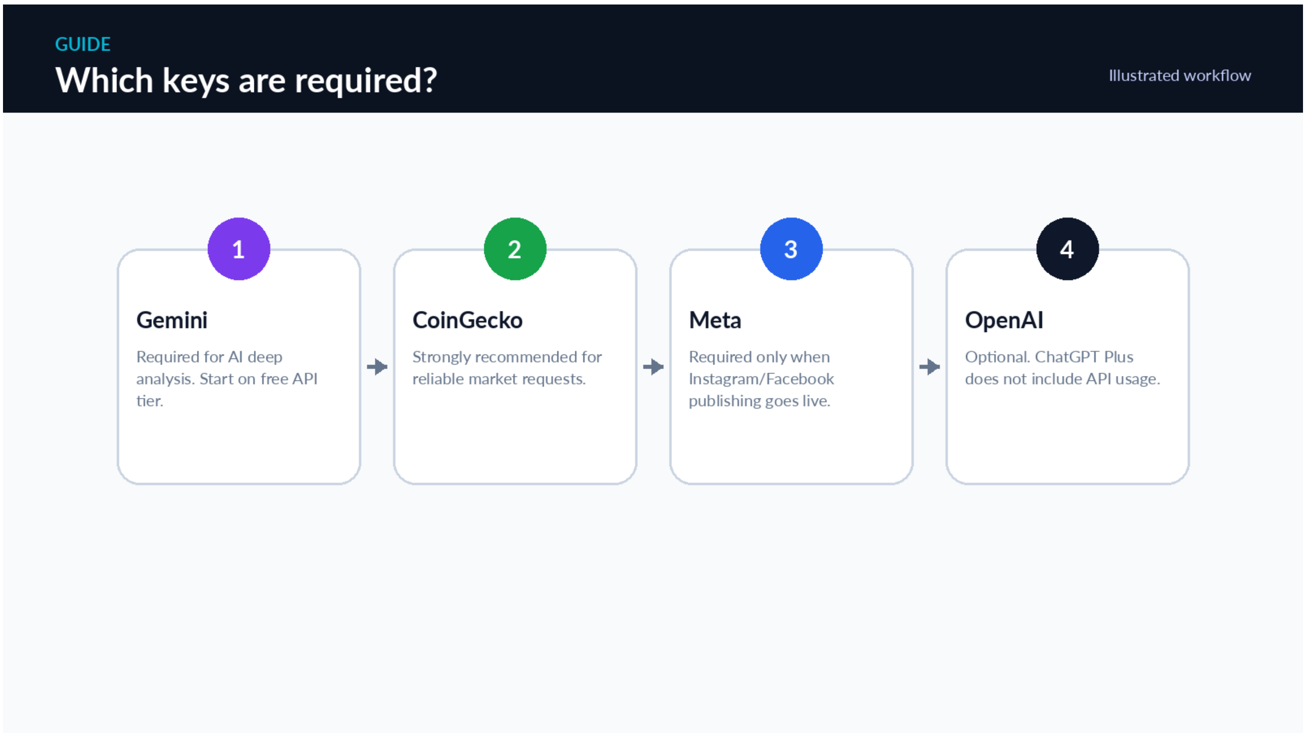
The deployment script clones the repository into /opt/cryptopulse, copies .env.example to .env if needed, builds the image and starts the app and Caddy containers.

8.4 Create the temporary public hostname

For the initial zero-cost deployment, sslip.io can map a hostname containing your public IP back to that IP. For example, public IP 129.153.42.10 becomes 129-153-42-10.sslip.io. This is suitable for testing, but a domain you control is better for long-term production. [S4]

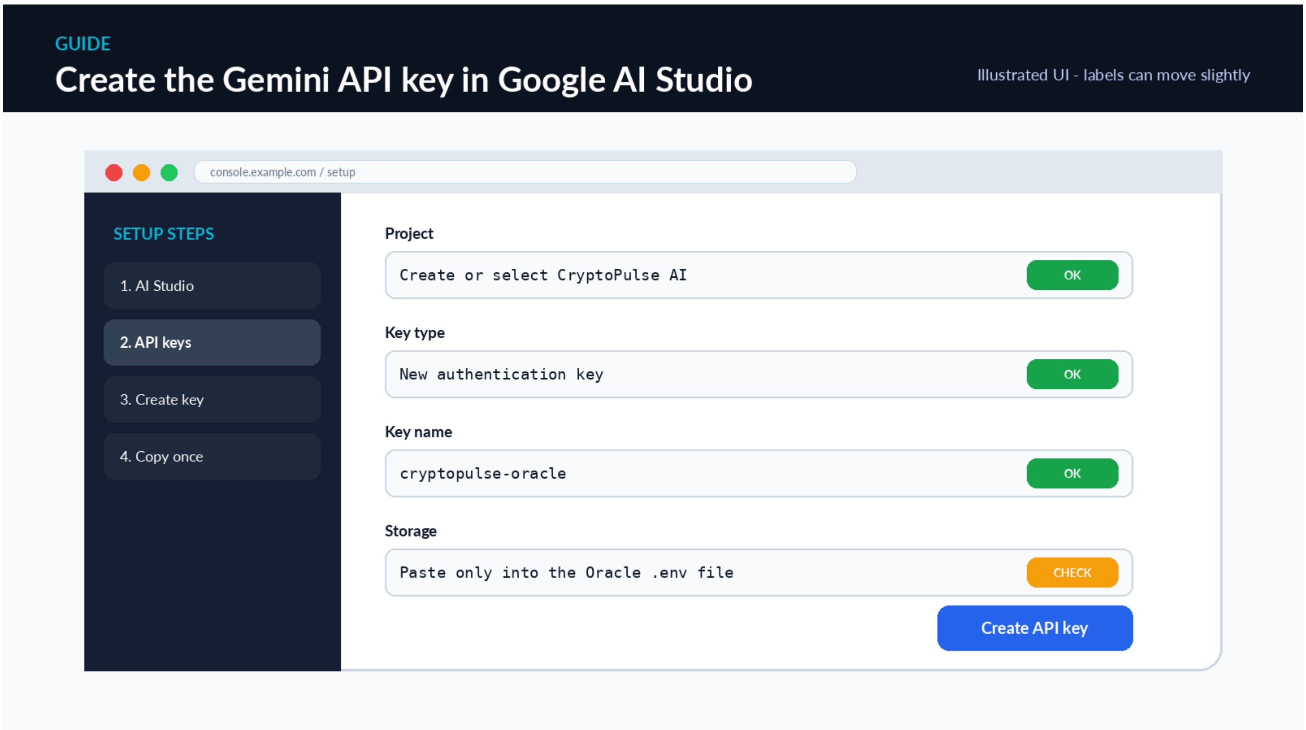
OCI public IP	Temporary hostname
129.153.42.10	129-153-42-10.sslip.io
YOUR_PUBLIC_IP	YOUR-IP-WITH-HYPHENS.sslip.io

9. Create and add the Gemini API key



> **Subscription distinction -** The Gemini consumer subscription and Gemini API are separate surfaces. Your cloud application needs an API key from Google AI Studio and remains subject to API quotas and billing rules.

9.1 Create the key



1	Open Google AI Studio and sign in with your Google account.
2	Open the API keys page.

3	New users may already have a project and key. To create a separate production key, click Create API key.
4	Create or select a project named CryptoPulse AI.
5	Create the key and copy it immediately.
6	Store the key temporarily in a password manager, not a text message or screenshot.

Google currently documents that new users can receive a project and key automatically, and new keys can be created from the AI Studio API keys page. [S5]

> **Key type change in 2026 -** Google documents a migration to authentication keys and restrictions for dormant unrestricted keys. Create a new key in AI Studio and use the key shown there. [S6]

9.2 Add it to Oracle

ORACLE TERMINAL

```
cd /opt/cryptopulse
nano .env
```

Find these lines and change them:

```
DRY_RUN=true
LLM_PROVIDER=gemini
GEMINI_API_KEY=PASTE_YOUR_REAL_GEMINI_KEY
GEMINI_MODEL=gemini-3.5-flash
LLM_MAX_CONCURRENCY=2
```

In nano: press Ctrl+O, press Enter to save, then Ctrl+X to exit.

```
chmod 600 .env
docker compose up -d --force-recreate app
docker compose exec app cryptopulse doctor
```

Gemini API keys

https://aistudio.google.com/apikey

Scan to open the official Google AI Studio API keys page.

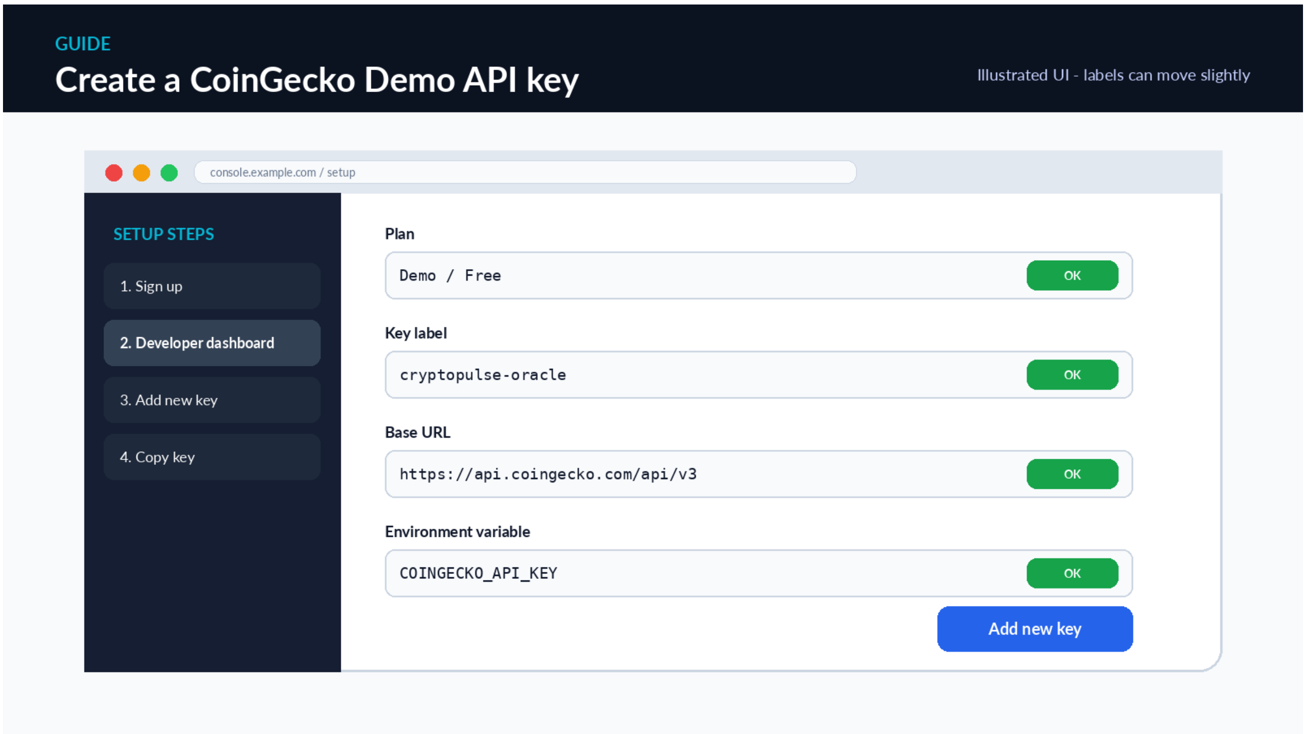
9.3 Test Gemini without publishing

```
docker compose exec app cryptopulse run-once
docker compose logs --since=30m app
```

Expected: the run completes in dry-run mode, creates media files and records dry-run publication IDs. If you see 429 or rate-limit errors, keep DRY_RUN=true and reduce LLM_MAX_CONCURRENCY to 1.

10. Create and add the CoinGecko API key

CoinGecko provides a Demo API key flow from the Developer Dashboard. Rate limits and monthly credits can change by plan, so check the Developer Dashboard and official API usage documentation before production. [S7] [S8]



10.1 Create the Demo key

1	Open CoinGecko API and create an account.
2	Choose the Demo / Free plan.
3	Open the Developer Dashboard.
4	Click + Add New Key.
5	Use label cryptopulse-oracle.
6	Copy the key and store it in your password manager.

10.2 Add it to .env

```
cd /opt/cryptopulse
nano .env
```

```
MARKET_PROVIDER=coingecko
COINGECKO_API_KEY=PASTE_YOUR_REAL_COINGECKO_KEY
COINGECKO_BASE_URL=https://api.coingecko.com/api/v3
COINGECKO_REQUEST_INTERVAL_SECONDS=2.2
```

```
chmod 600 .env
docker compose up -d --force-recreate app
docker compose exec app cryptopulse doctor
docker compose exec app cryptopulse run-once
```

Do not use the Pro URL - A Demo key must use api.coingecko.com. CoinGecko documents that using the wrong base URL causes an invalid-key error. [S8]

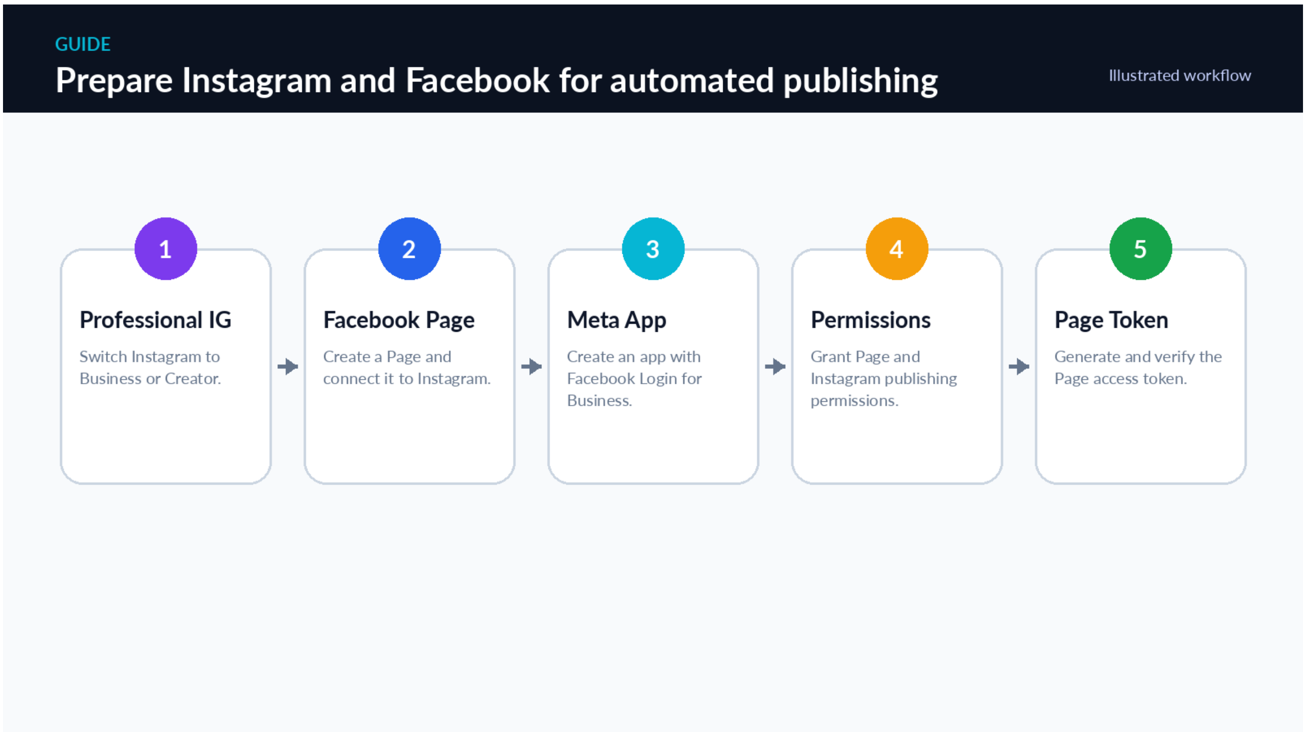
CoinGecko API

https://www.coingecko.com/en/api

Scan to open CoinGecko API information.

11. Prepare Meta, Instagram and Facebook publishing

Most complex chapter - Complete every step while DRY_RUN=true. Meta UI and permission labels can change; use the current official documentation linked in the Sources chapter when a screen differs.



11.1 Prepare Instagram

1	Open Instagram and go to Settings and privacy.
2	Open Account type and tools or the equivalent professional-account menu.
3	Switch to a Professional account.
4	Choose Business or Creator. Business is usually clearer for a publishing brand.
5	Complete the category and public profile information.

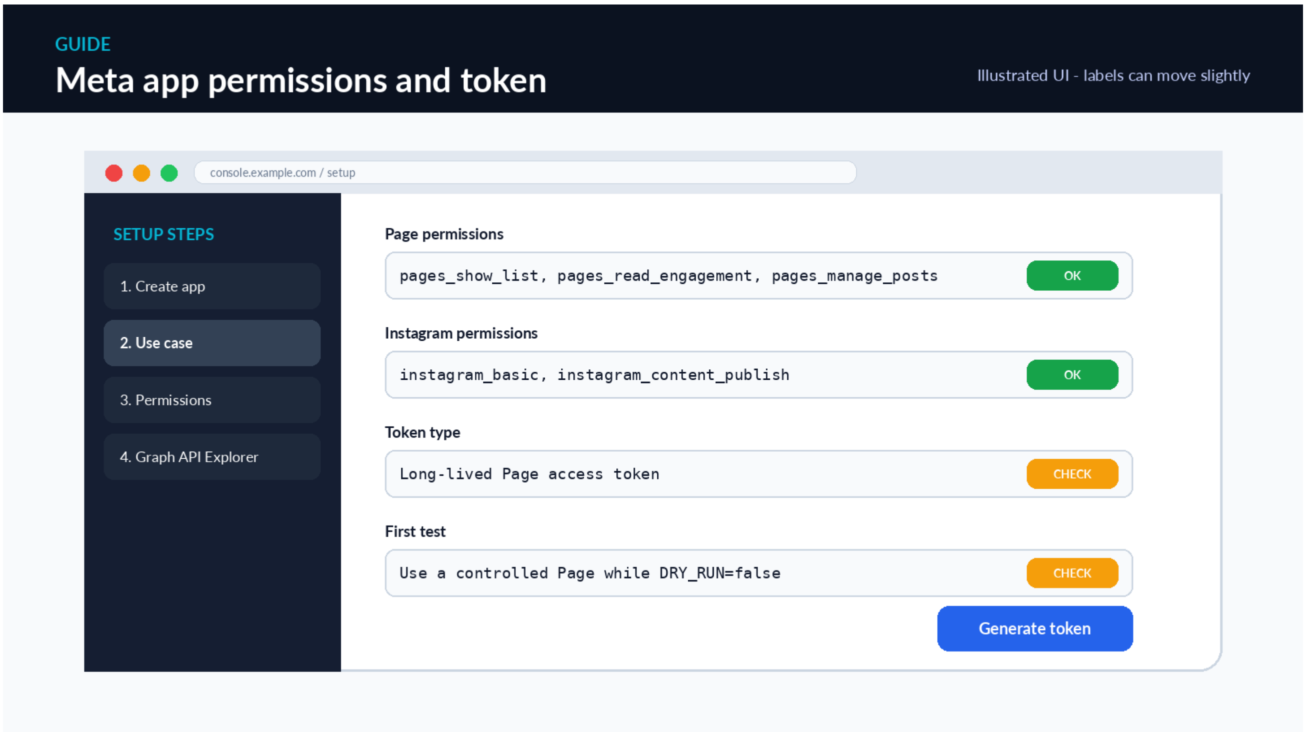
11.2 Create and connect the Facebook Page

1	Create a Facebook Page for the CryptoPulse brand if you do not already have one.
2	Ensure your Facebook profile has full control/admin access to the Page.
3	In Page or Meta Business settings, connect the Instagram professional account to the Page.

4	Confirm both accounts appear under the same business portfolio or connected-assets screen.

The current Meta platform supports Instagram professional accounts. The repository uses a Page-based Facebook Login for Business flow because it must publish to both Instagram and the Facebook Page. [S9] [S10]

11.3 Create the Meta developer app



1	Open Meta for Developers and create a developer account if requested.
2	Click My Apps, then Create App.
3	Choose the business/content management use case available for Facebook and Instagram publishing.
4	Add Facebook Login for Business and Instagram API products when prompted.
5	Use your own business portfolio, Facebook Page and connected Instagram professional account.

11.4 Required permissions

Purpose	Permissions used by this repository
List and identify the Page	pages_show_list
Read Page engagement required by APIs	pages_read_engagement
Publish Facebook Page posts/photos	pages_manage_posts
Read connected Instagram account	instagram_basic
Publish Instagram content	instagram_content_publish

Meta currently documents Page permissions including pages_show_list, pages_read_engagement and pages_manage_posts, and Instagram publishing permissions depending on login type. The Facebook Login path uses instagram_basic and instagram_content_publish; the Instagram Login path uses newer instagram_business_* names. This project is configured for the Page/Facebook Login path so one Page access token can serve both publishers. [S10] [S11]

11.5 Generate a Page access token

1	Open Graph API Explorer in Meta for Developers.
2	Select your CryptoPulse app.
3	Generate a User Access Token with the required Page and Instagram permissions.
4	Run GET /me/accounts?fields=id,name,access_token,instagram_business_account.
5	Find your Page in the response.
6	Copy the Page id into FACEBOOK_PAGE_ID.
7	Copy instagram_business_account.id into INSTAGRAM_USER_ID.
8	Copy the Page access_token into META_ACCESS_TOKEN.
9	Use Meta token tools to obtain and verify the appropriate long-lived token for production.

> **Token lifetime -** Meta documents that a long-lived Page token can be generated from a long-lived User token. Token behavior can be affected by account changes, password changes, app mode and permissions. Monitor and rotate it. [S12]

11.6 Test the IDs without exposing the token

```
cd /opt/cryptopulse
nano .env
```

```
META_GRAPH_VERSION=v25.0
META_ACCESS_TOKEN=PASTE_PAGE_ACCESS_TOKEN
INSTAGRAM_USER_ID=PASTE_NUMERIC_IG_ID
FACEBOOK_PAGE_ID=PASTE_NUMERIC_PAGE_ID
PUBLISH_INSTAGRAM=true
PUBLISH_FACEBOOK=true
PUBLISH_STORIES=true
PUBLISH_REELS=true
DRY_RUN=true
```

Meta Developers

https://developers.facebook.com

Scan to open Meta for Developers.

12. Complete the .env configuration

CONFIGURATION

Where each API key goes in .env

Never upload .env to GitHub

DRY_RUN	=	true	Keep true until all tests pass
LLM_PROVIDER	=	gemini	Choose gemini or heuristic
GEMINI_API_KEY	=	YOUR_GEMINI_KEY	Paste Gemini key here
COINGECKO_API_KEY	=	YOUR_COINGECKO_KEY	Paste Demo API key
META_ACCESS_TOKEN	=	YOUR_PAGE_ACCESS_TOKEN	Required only for live publishing
INSTAGRAM_USER_ID	=	1784XXXXXXXXXXX	Numeric professional account ID
FACEBOOK_PAGE_ID	=	1234XXXXXXXXXXX	Numeric Page ID
PUBLIC_HOST	=	129-153-42-10.sslip.io	Use your OCI public IP
MEDIA_SIGNING_SECRET	=	LONG_RANDOM_SECRET	Generate with supplied script

12.1 Generate the media signing secret

```
cd /opt/cryptopulse
bash scripts/generate_secret.sh
```

Copy the printed random value and place it after MEDIA_SIGNING_SECRET= in .env.

12.2 Build the temporary HTTPS host

Example only: if your OCI public IP is 129.153.42.10, use 129-153-42-10.sslip.io.

```
PUBLIC_HOST=YOUR-IP-WITH-HYPHENS.sslip.io
PUBLIC_BASE_URL=https://YOUR-IP-WITH-HYPHENS.sslip.io
MEDIA_SIGNING_SECRET=PASTE_GENERATED_SECRET
```

12.3 Recommended first complete .env

```
# Runtime
APP_ENV=production
LOG_LEVEL=INFO
DRY_RUN=true
TIMEZONE=UTC
SCHEDULE_HOURS=0,6,12,18
RUN_ON_STARTUP=false
MAX_CONCURRENT_WORKERS=4
MAX_MARKET_PAGES=80
MARKET_PAGE_SIZE=250
MIN_MARKET_CAP_USD=1000000
MIN_VOLUME_24H_USD=100000
TOP_PRELIMINARY_PER_SIDE=15
TOP_FINAL_PER_SIDE=5
PREDICTION_HORIZON_HOURS=6

# Data providers
MARKET_PROVIDER=coingecko
COINGECKO_API_KEY=
```

```
COINGECKO_BASE_URL=https://api.coingecko.com/api/v3
COINGECKO_REQUEST_INTERVAL_SECONDS=2.2
NEWS_PROVIDER=hybrid
NEWS_RSS_FEEDS=https://www.coindesk.com/arc/outboundfeeds/rss/,https://cointelegraph.com/
rss,https://decrypt.co/feed
GDELT_MAX_RECORDS=75

# AI provider. Use "gemini", "openai", or "heuristic".
LLM_PROVIDER=gemini
GEMINI_API_KEY=
GEMINI_MODEL=gemini-3.5-flash
OPENAI_API_KEY=
OPENAI_MODEL=gpt-5-mini
LLM_BATCH_SIZE=10
LLM_MAX_CONCURRENCY=2
LLM_TIMEOUT_SECONDS=90

# Meta publishing
META_GRAPH_VERSION=v25.0
META_ACCESS_TOKEN=
INSTAGRAM_USER_ID=
FACEBOOK_PAGE_ID=
PUBLISH_INSTAGRAM=true
PUBLISH_FACEBOOK=true
PUBLISH_STORIES=true
PUBLISH_REELS=true

# Public HTTPS media serving. Example host: 203-0-113-10.sslip.io
PUBLIC_HOST=localhost
PUBLIC_BASE_URL=
MEDIA_SIGNING_SECRET=change-this-to-a-long-random-secret
MEDIA_URL_TTL_SECONDS=86400
MEDIA_DELETE_GRACE_HOURS=24

# Database and storage
DATABASE_URL=sqlite:////app/data/cryptopulse.db
MEDIA_ROOT=/app/data/media
RETAIN_RAW_DATA=false
RETAIN_GENERATED_MEDIA=false
PROJECT_PROFILE_REFRESH_HOURS=168

# Branding
BRAND_NAME=CryptoPulse AI
BRAND_HANDLE=@yourhandle
DISCLAIMER=Educational market intelligence only. Not financial advice. Crypto assets are highly
volatile.
```

Edit only what is necessary - For the first dry run, keep DRY_RUN=true, RETAIN_GENERATED_MEDIA=true and RUN_ON_STARTUP=false. This lets you inspect the output and prevents automatic live publishing.

12.4 Apply configuration and start services

```
cd /opt/cryptopulse
chmod 600 .env
docker compose config
docker compose build --pull
docker compose up -d --remove-orphans
docker compose ps
```

Expected: app and caddy show Up. The app may take 20-60 seconds to become healthy on the first start.

13. Run tests in dry-run mode

13.1 Health checks

```
cd /opt/cryptopulse
curl http://127.0.0.1:8080/health
curl http://127.0.0.1:8080/ready
docker compose exec app cryptopulse doctor
```

Check	Pass condition
/health	HTTP 200 and healthy response
/ready	HTTP 200 after providers/configuration are ready
cryptopulse doctor	No missing required configuration for current mode
docker compose ps	app and caddy running; app healthy

13.2 Full smoke test

```
bash scripts/smoke_test.sh
```

Because DRY_RUN=true, the smoke script is allowed to run a complete cycle without publishing real content.

13.3 Manual run

```
docker compose exec app cryptopulse run-once
```

The first real-provider run can take several minutes because the system pages market data, scans news, calls Gemini and renders media. Do not start a second run while the first run is active.

13.4 View logs

```
docker compose logs --since=30m app
docker compose logs -f app
```

Press Ctrl+C to leave the live log view. Ctrl+C here stops only log viewing; it does not stop the containers.

13.5 Confirm database and media

```
docker compose exec app sh -lc "find /app/data/media -maxdepth 3 -type f | sort | tail -40"
docker compose exec app sh -lc "ls -lh /app/data/cryptopulse.db"
```

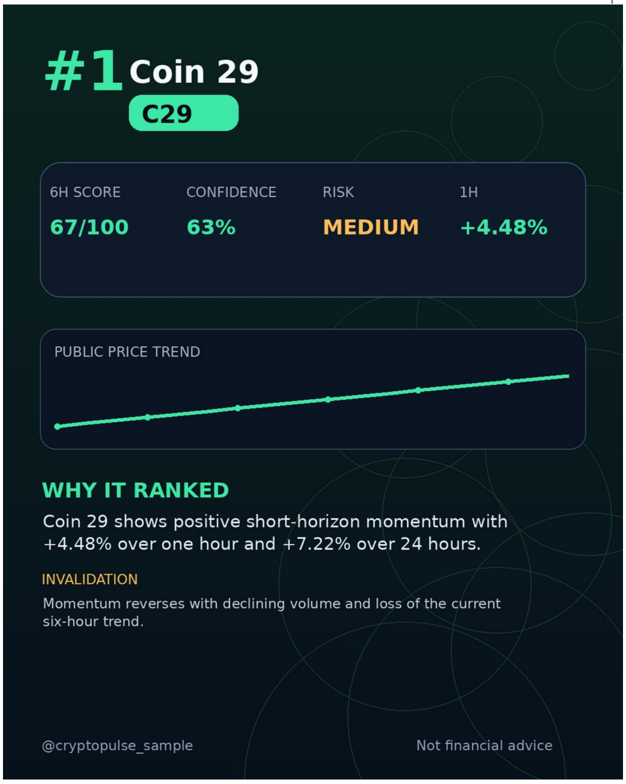
Do not enable live mode yet - A successful dry-run proves the internal workflow. It does not prove that Meta can download your files or that the access token can publish.

14. Review generated images and public HTTPS

14.1 What the output should look like

Example Bullish Watch carousel cover.

Example Bearish Risk carousel cover.



Example ranked asset slide.

- Correct Bullish Watch or Bearish Risk direction.
- Exactly five ranked assets after the cover.
- No spelling mistakes or text outside the image.
- Confidence and risk labels are readable.
- Timestamp and six-hour horizon are present where configured.
- Disclaimer says educational market intelligence, not financial advice.

14.2 Test public HTTPS

```
curl -I https://YOUR-IP-WITH-HYPHENS.sslip.io/health
curl https://YOUR-IP-WITH-HYPHENS.sslip.io/health
```

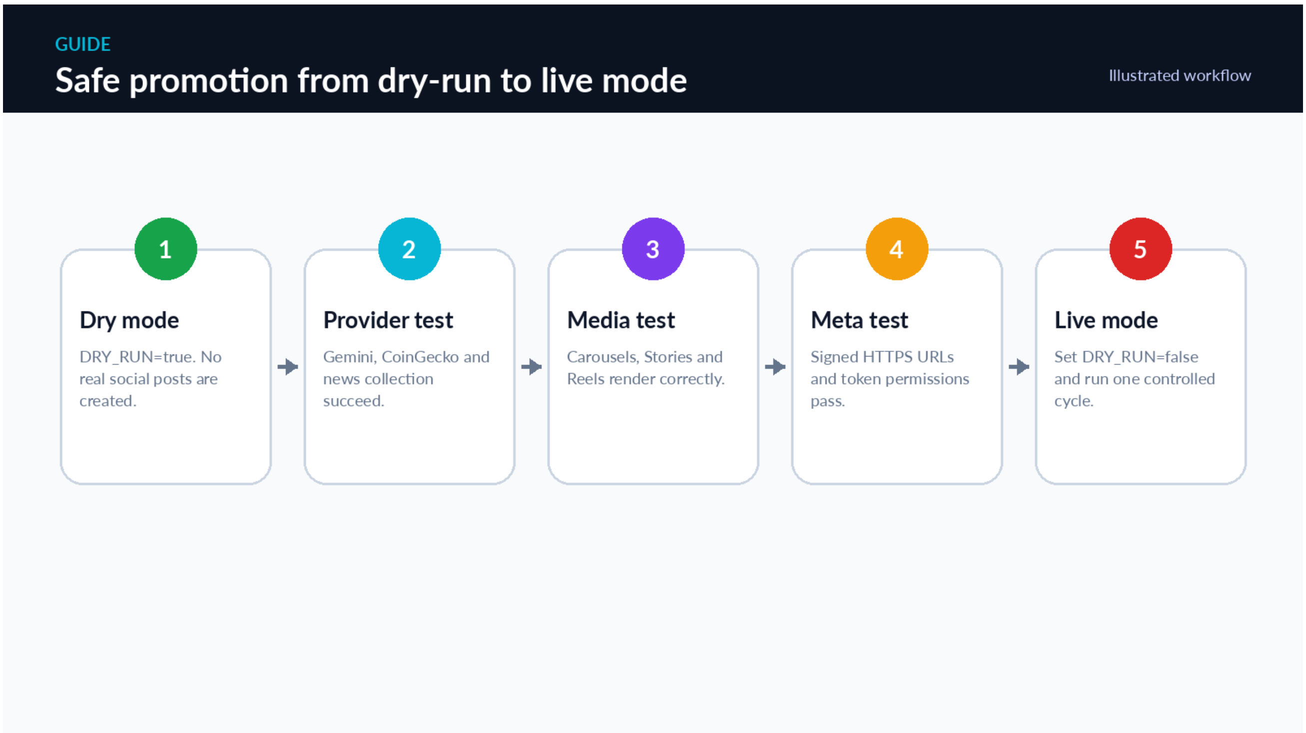
Then turn off Wi-Fi on your phone, use mobile data, and open the same HTTPS health URL. This confirms the service is reachable from outside Oracle.

14.3 Test a signed media URL

The application creates signed, expiring URLs for media. Run one dry cycle, find the signed URL in logs or run summary, and open it from mobile data. Meta must be able to download that exact URL without authentication.

> **Caddy certificate problem -** Verify ports 80 and 443 are open in both OCI security rules and UFW. Confirm PUBLIC_HOST contains the correct current public IP.

15. Enable one controlled live post



15.1 Final pre-live checklist

- GitHub Actions is green for the deployed commit.
- docker compose ps shows app and caddy running.
- /health and /ready return healthy responses.
- Gemini and CoinGecko dry run completed.
- Generated carousels, Stories and Reel were inspected.
- Public HTTPS health and signed media URLs work from mobile data.
- Meta Page token, FACEBOOK_PAGE_ID and INSTAGRAM_USER_ID were verified.
- The Instagram account and Facebook Page are controlled test/production assets.
- A database backup was created.

15.2 Create a backup before first live post

```
cd /opt/cryptopulse
bash scripts/backup_db.sh
ls -lh backups
```

15.3 Switch to live mode

```
nano /opt/cryptopulse/.env
```

```
DRY_RUN=false
RETAIN_GENERATED_MEDIA=true
RUN_ON_STARTUP=false
```

Keep RETAIN_GENERATED_MEDIA=true for the first live test so you can diagnose any publishing problem. Change it to false only after reliable operation.

```
cd /opt/cryptopulse
docker compose up -d --force-recreate app
docker compose exec app cryptopulse doctor
```

15.4 Run exactly one live cycle

```
docker compose exec app cryptopulse run-once
docker compose logs --since=30m app
```

1	Wait for the command to finish.
2	Open Instagram and Facebook and inspect every published item.
3	Confirm no duplicate posts were created.
4	Confirm external Meta post IDs appear in logs or records.
5	If anything is wrong, immediately set DRY_RUN=true, recreate the app container and investigate.

15.5 Enable normal cleanup and scheduling

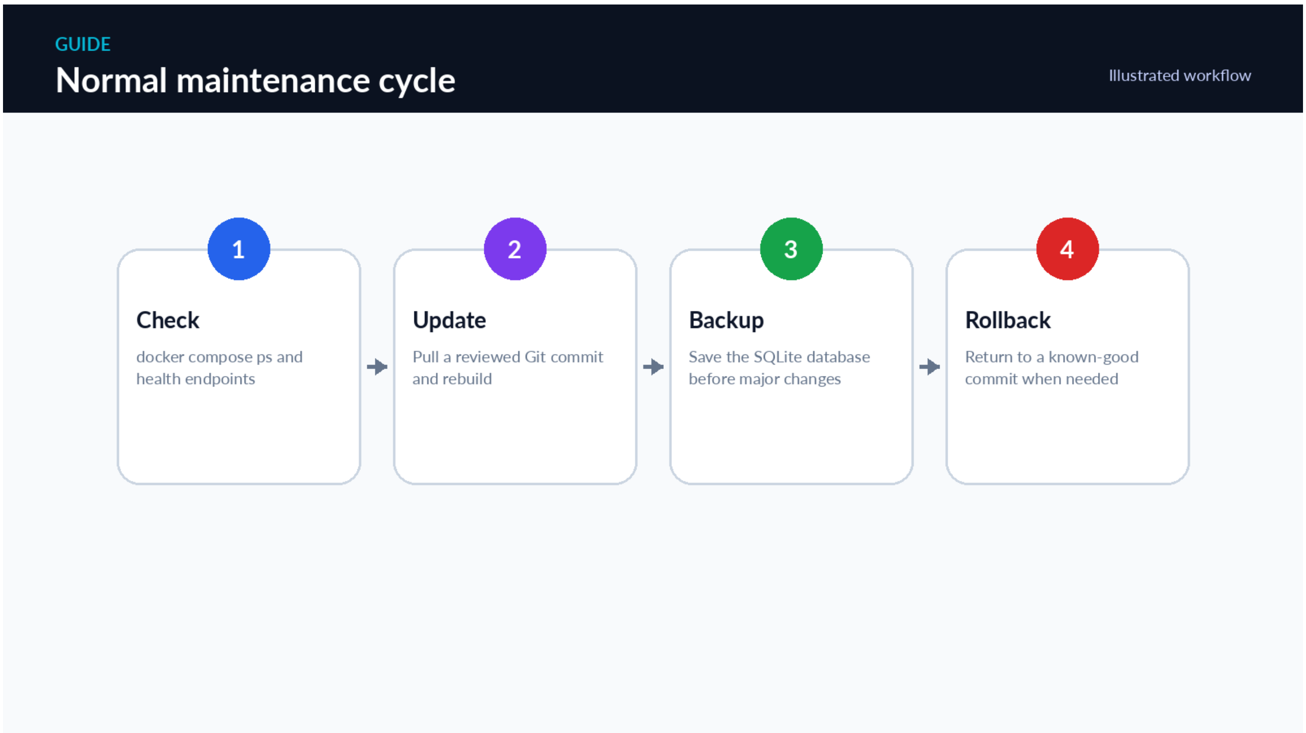
```
nano /opt/cryptopulse/.env
```

```
DRY_RUN=false
RETAIN_GENERATED_MEDIA=false
MEDIA_DELETE_GRACE_HOURS=24
RUN_ON_STARTUP=false
SCHEDULE_HOURS=0,6,12,18
```

```
docker compose up -d --force-recreate app
docker compose ps
```

Financial communication - Keep the wording Bullish Watch and Bearish Risk. Do not change it to guaranteed profit, buy now or sell now language.

16. Daily operations, updates, backups and rollback



16.1 Health check commands

```
cd /opt/cryptopulse
docker compose ps
curl https://$PUBLIC_HOST/health
curl https://$PUBLIC_HOST/ready
docker compose logs --since=30m app
```

16.2 Update from Git

Push reviewed changes to GitHub first. Then run:

```
cd /opt/cryptopulse
bash scripts/update_from_git.sh main
```

The script pulls the branch, rebuilds the image, restarts services and performs a health check.

16.3 Create a manual database backup

```
cd /opt/cryptopulse
bash scripts/backup_db.sh
ls -lh backups
```

16.4 Find a known-good commit

```
cd /opt/cryptopulse
git log --oneline -10
```

16.5 Roll back

```
cd /opt/cryptopulse
bash scripts/rollback.sh KNOWN_GOOD_COMMIT_SHA
```

Rollback mode - The rollback script checks out a detached commit. After recovery, create a proper fix in Git and redeploy the main branch.

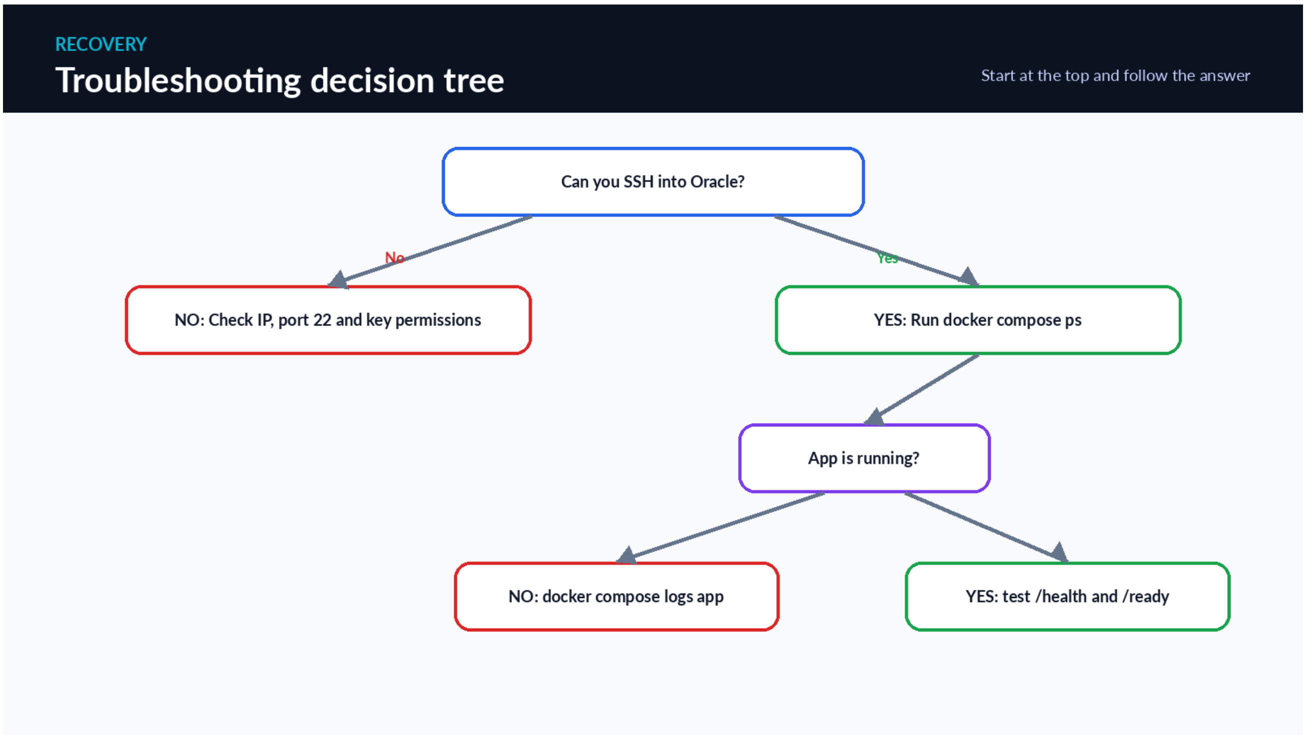
16.6 Stop and restart

```
cd /opt/cryptopulse
docker compose down
docker compose up -d
docker compose restart app
```

16.7 Check disk usage

```
df -h
docker system df
du -sh /opt/cryptopulse/backups
```

17. Troubleshooting



17.1 SSH errors

Error	Likely cause	Fix
Connection timed out	Port 22, wrong IP, instance stopped	Check OCI state, public IP and ingress rule.
Permission denied (publickey)	Wrong key/username/permissions	Use ubuntu and correct .key; secure key permissions.
Host key changed	Instance was recreated or IP reused	Verify instance identity, then remove old known_hosts entry.

17.2 Docker errors

```
cd /opt/cryptopulse
docker compose ps
docker compose logs --tail=200 app
docker compose logs --tail=200 caddy
docker info
```

Symptom	Fix
permission denied /var/run/docker.sock	Log out and back in after bootstrap; confirm groups command includes docker.
container exits immediately	Read app logs; validate .env with docker compose config.
image build fails	Check internet, disk space and ARM-compatible dependencies.
app unhealthy	Wait 60 seconds, then inspect /health and logs.

17.3 Gemini errors

Error	Action
401 / invalid key	Create a new AI Studio key and update GEMINI_API_KEY.
403 / blocked key	Create a current authentication key and review Google key status.
429 / rate limit	Set LLM_MAX_CONCURRENCY=1 and retry later.
timeout	Increase LLM_TIMEOUT_SECONDS cautiously and check

	connectivity.

17.4 CoinGecko errors

Error	Action
10002 missing key	Add COINGECKO_API_KEY.
10010/10011 wrong key type	Demo key must use api.coingecko.com.
429 rate limit	Keep request interval, wait, and reduce market pages during testing.
empty market result	Check network, provider status and filtering thresholds.

17.5 Meta errors

Symptom	Action
Invalid OAuth token	Regenerate/extend token and update .env.
Unsupported request or permission	Review token scopes and Page/IG asset access.
Media fetch failed	Open signed URL from mobile data; verify HTTPS and file retention.
Container stuck	Inspect status logs; keep media for longer and retry carefully.
Facebook works, Instagram fails	Verify connected professional account and INSTAGRAM_USER_ID.
Duplicate post concern	Do not re-run blindly; inspect Meta and stored publish IDs first.

17.6 Emergency safe mode

```
cd /opt/cryptopulse
sed -i "s/^DRY_RUN=.*/DRY_RUN=true/" .env
docker compose up -d --force-recreate app
docker compose ps
```

Use safe mode first - When uncertain, stop live publishing by restoring DRY_RUN=true before attempting repairs.

18. Security checklist and final acceptance

SECURITY

API key security: do this / never do this

Treat every key like a bank password

DO

- Store keys only in /opt/cryptopulse/.env
- Run chmod 600 .env
- Use separate keys per environment
- Delete and replace a leaked key
- Keep DRY_RUN=true during setup

NEVER

- Put keys in README or source code
- Commit .env to GitHub
- Send keys in WhatsApp or email
- Show full keys in screenshots
- Paste keys into unknown websites

18.1 Final security checklist

- Oracle private SSH key is stored in your Windows .ssh folder and backed up securely.
- GitHub repository is private and uses a read-only Oracle deploy key.
- .env is not tracked by Git and has chmod 600 permissions on Oracle.
- No API key appears in README, source files, logs, screenshots or chat messages.
- Only ports 22, 80 and 443 are open; 8080 and database ports are not public.
- All accounts use multi-factor authentication where available.
- OpenAI is disabled unless separate API billing and a separate API key are intentionally configured.
- Backups are created before upgrades or live-mode changes.
- Live financial content always contains a disclaimer and probabilistic language.

18.2 Production acceptance record

Acceptance item	Result / date / note
GitHub CI passed	____
Oracle smoke test passed	____
Gemini dry run passed	____
CoinGecko dry run passed	____
Public HTTPS passed from mobile data	____
Signed media URL passed	____
Meta controlled live post passed	____
Backup completed	____
Scheduled live mode enabled	____

> **Honest limitation -** No software can be guaranteed to have zero errors under every external API, network, platform-policy or market condition. This workflow reduces risk through tests, dry-run gates, bounded retries, audit records, backups and rollback.

Appendix A - Command quick reference

Windows PowerShell

```
# Connect to Oracle
ssh -i "$HOME\.ssh\cryptopulse_oci.key" ubuntu@YOUR_PUBLIC_IP

# Push local changes
cd "$HOME\Downloads\cryptopulse_release\cryptopulse_ai"
git add .
git commit -m "Describe the change"
git push
```

Oracle: normal operations

```
cd /opt/cryptopulse
docker compose ps
docker compose logs --since=30m app
docker compose exec app cryptopulse doctor
docker compose exec app cryptopulse run-once
bash scripts/backup_db.sh
bash scripts/update_from_git.sh main
```

Oracle: safe recovery

```
cd /opt/cryptopulse
sed -i "s/^DRY_RUN=.*/DRY_RUN=true/" .env
docker compose up -d --force-recreate app
docker compose logs --tail=200 app
git log --oneline -10
bash scripts/rollback.sh KNOWN_GOOD_COMMIT_SHA
```

Appendix B - API and account worksheet

Item	Value / location (never write full secret in a printed copy)
GitHub username	____
Repository	git@github.com:_______________/cryptopulse-ai.git
OCI public IP	____
PUBLIC_HOST	__________________________________.sslip.io
Gemini project/key name	____
CoinGecko key label	____
Meta App ID	____
Facebook Page ID	____
Instagram User ID	____
Token rotation reminder date	____

Sources and official documentation

Platform interfaces, permissions, model names and quotas can change. These official sources were checked on 20 July 2026. Use them whenever a current screen differs from an illustration in this manual.

[S1] Oracle - Sign Up for the Free Oracle Cloud Promotion: https://docs.oracle.com/en-us/iaas/Content/GSG/Tasks/signingup_topic-Sign_Up_for_Free_Oracle_Cloud_Promotion.htm

[S2] Oracle - Always Free Resources: https://docs.oracle.com/en-us/iaas/Content/FreeTier/freetier_topic-Always_Free_Resources.htm

[S3] Oracle - Connecting to a Linux Instance from Windows: https://docs.oracle.com/en-us/iaas/Content/Compute/Tasks/connect-to-linux-instance.htm

[S4] sslip.io - Wildcard DNS service: https://sslip.io/

[S5] Google - Gemini API Getting Started: https://ai.google.dev/gemini-api/docs/get-started

[S6] Google - Using Gemini API keys: https://ai.google.dev/gemini-api/docs/api-key

[S7] CoinGecko - Setting Up Your API Key: https://docs.coingecko.com/docs/setting-up-your-api-key

[S8] CoinGecko - Errors and Rate Limits: https://docs.coingecko.com/docs/errors-and-rate-limits

[S9] Meta - Instagram Platform Overview: https://developers.facebook.com/documentation/instagram-platform/overview

[S10] Meta - Facebook Pages API: https://developers.facebook.com/documentation/pages-api

[S11] Meta - Instagram Content Publishing: https://developers.facebook.com/documentation/instagram-platform/content-publishing

[S12] Meta - Long-Lived Access Tokens: https://developers.facebook.com/documentation/facebook-login/guides/access-tokens/get-long-lived

[S13] GitHub - Adding locally hosted code to GitHub: https://docs.github.com/en/migrations/importing-source-code/using-the-command-line-to-import-source-code/adding-locally-hosted-code-to-github

[S14] GitHub - Generating and adding an SSH key: https://docs.github.com/en/authentication/connecting-to-github-with-ssh/generating-a-new-ssh-key-and-adding-it-to-the-ssh-agent

[S15] OpenAI - ChatGPT Plus does not include API usage: https://help.openai.com/en/articles/6950777-what-is-chatgpt-plus

[S16] OpenAI - Find and create API keys: https://help.openai.com/en/articles/4936850-where-do-i-find-my-openai-api-key

[S17] OpenAI - API key safety: https://help.openai.com/en/articles/5112595-best-practices-for-api-key-safety

Optional OpenAI configuration

ChatGPT Plus does not include OpenAI API usage; API billing is separate. [S15] The project can run on Gemini only. Configure OpenAI only after intentionally creating a separate API project/key and payment controls.

```
LLM_PROVIDER=openai
OPENAI_API_KEY=PASTE_SEPARATE_OPENAI_API_KEY
OPENAI_MODEL=gpt-5-mini
```

Optional means optional - Do not add an OpenAI key merely because you have ChatGPT Plus. Gemini is the selected zero-budget starting provider.

YOU ARE READY TO START

Use the acceptance checklist, keep dry-run enabled until every gate passes, and change one setting at a time.

CryptoPulse AI - Institutional workflow, beginner-safe deployment

PART II

AI Inbox and Promotion Manager

Application v0.2.0 setup, testing and operations

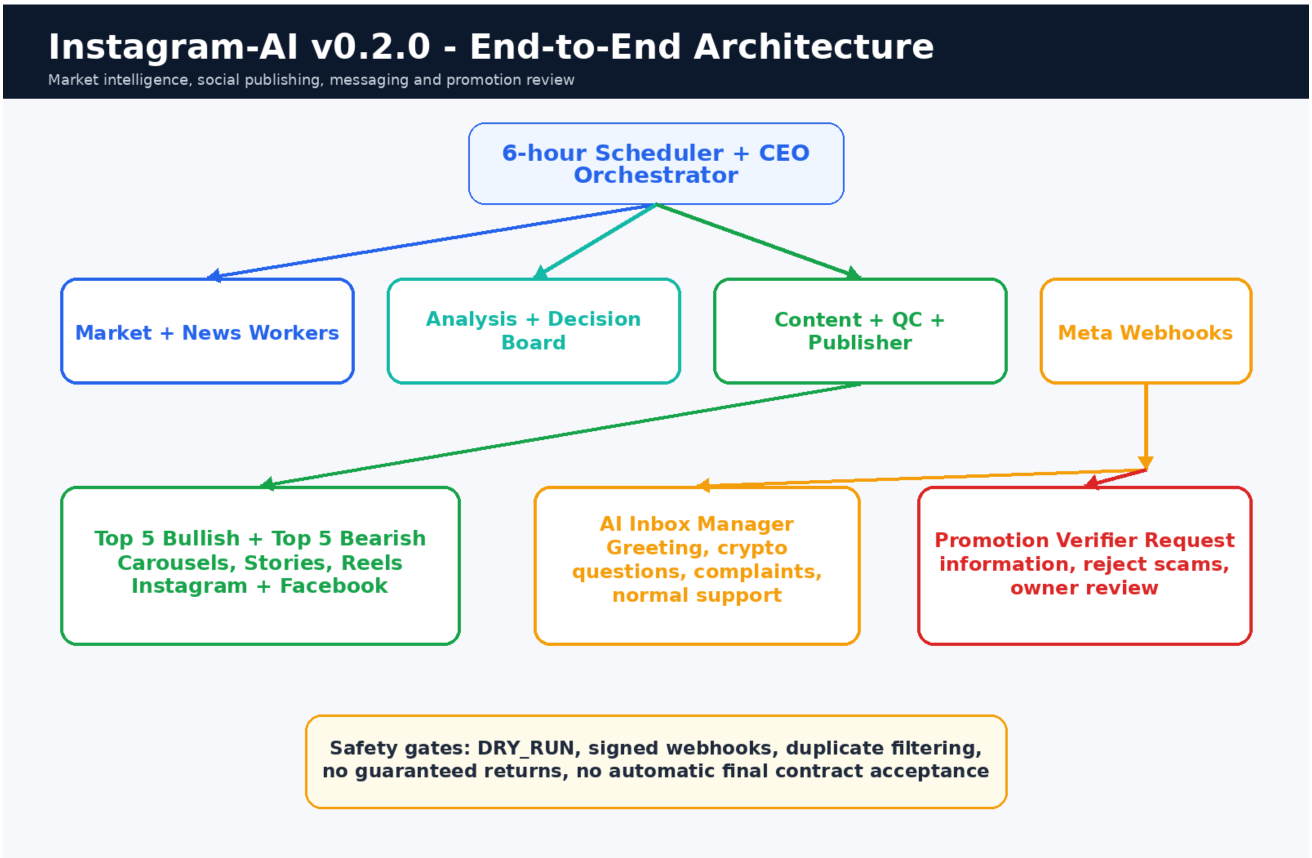


Figure 19-1. Complete v0.2.0 architecture.

Important release boundary

The market-analysis, media-generation and webhook pipelines passed offline automated tests. Live Gemini, Meta publishing and real Instagram/Facebook messaging still require your private credentials, account permissions and controlled canary tests.

19. Version 0.2.0 requirements and feature status

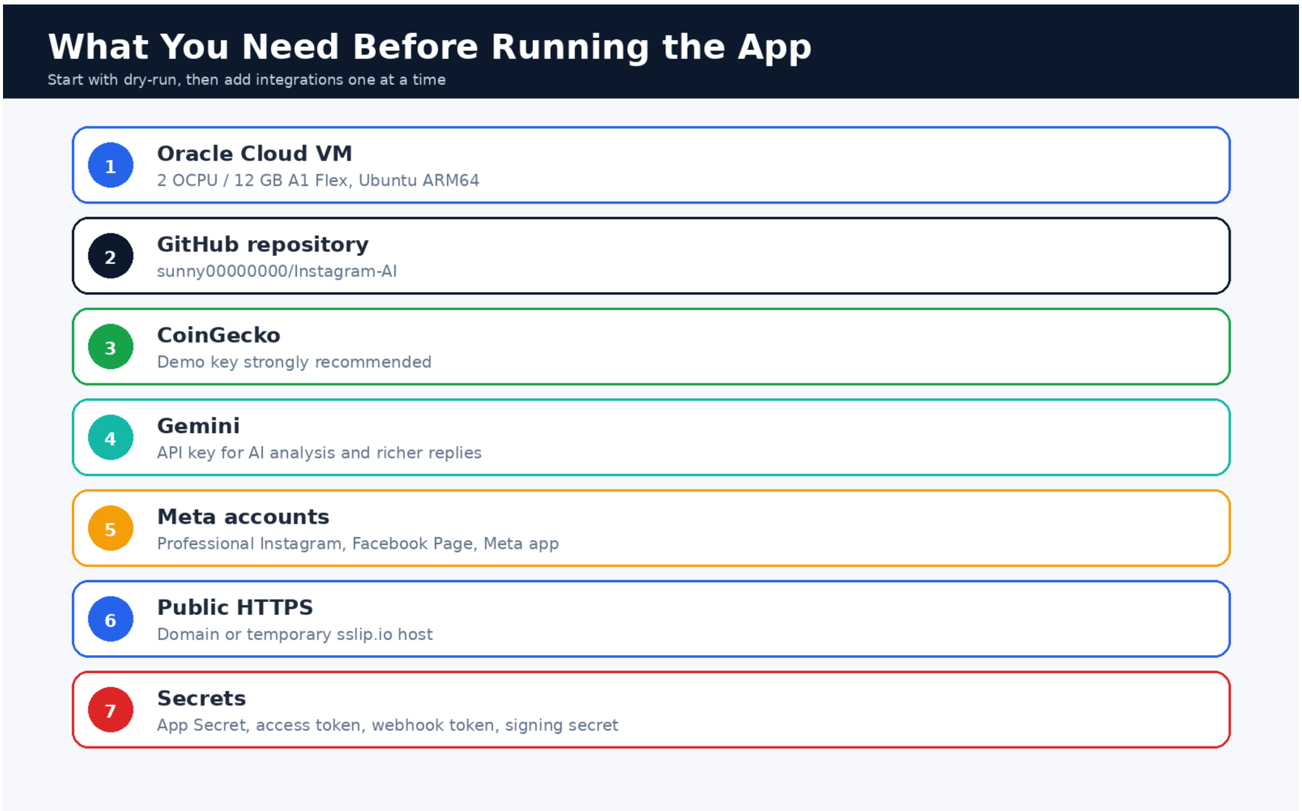


Figure 19-2. Accounts, infrastructure and credentials required for full operation.

19.1 Minimum hardware and accounts

Requirement	Minimum / recommended	Why it is needed
Oracle Cloud VM	VM.Standard.A1.Flex; 2 OCPU, 12 GB RAM; Ubuntu ARM64; 50-100 GB boot volume	Runs Docker, scheduler, rendering, API, webhooks and Caddy.
GitHub	Repository sunny00000000/Instagram-AI	Source of truth for deployment and updates.
Public IP + HTTPS	Ports 22, 80 and 443 open; domain or sslip.io hostname	Meta must reach webhooks and download generated media.
CoinGecko	Demo key strongly recommended	Market prices, rankings and historical data.
Gemini	API key recommended	AI market analysis and caption quality. Inbox v0.2.0 itself uses safe templates.
Meta assets	Professional Instagram account, Facebook Page when applicable, Meta developer app	Publishing, tokens, IDs and messaging webhooks.
Windows PC	Browser, PowerShell/OpenSSH, downloaded Oracle SSH key	Initial account setup and server administration.

19.2 What works without external keys

- Application installation, Docker build, health endpoints and automated tests.
- Heuristic market analysis by setting LLM_PROVIDER=heuristic.
- Sample image generation and dry-run publishing records.
- Webhook verification and message-processing tests using simulated signed payloads.

- Promotion classification tests without contacting real advertisers.

19.3 What needs private credentials

Feature	Required values	Safe initial switch
Gemini analysis	GEMINI_API_KEY	LLM_PROVIDER=heuristic until tested
Instagram/Facebook publishing	META_ACCESS_TOKEN, INSTAGRAM_USER_ID, FACEBOOK_PAGE_ID, public HTTPS	DRY_RUN=true and all PUBLISH_* false
Instagram/Facebook inbox	META_APP_SECRET, META_WEBHOOK_VERIFY_TOKEN, META_ACCESS_TOKEN, subscribed messaging permissions	AUTO_REPLY_ENABLED=false
Optional OpenAI	Separate API key and API billing	Leave OPENAI_API_KEY empty

> **ChatGPT Plus is not an API key**
> ChatGPT Plus and OpenAI API billing are separate. This application can operate with Gemini and does not require an OpenAI API key.

19.4 Exact feature status in v0.2.0

Capability	Status	Important meaning
Market and news pipeline	Implemented and offline-tested	Runs every six hours and produces ranked content.
Instagram/Facebook publishing	Implemented; live credentials required	Keep dry-run until a controlled real post succeeds.
Webhook signature verification	Implemented and tested	Invalid HMAC signatures return HTTP 403.
Duplicate/echo message protection	Implemented and tested	Repeated message IDs and app echo events are ignored.
Normal inbox replies	Implemented as deterministic safe templates	Replies are predictable; not yet free-form Gemini conversation.
Promotion verification	Initial text-based screening	Extracts website/email/budget and red flags; not independent legal or corporate due diligence.
Final advertising acceptance	Owner approval only	The app must never sign, pay, accept money or approve contracts automatically.
Conversation retention	Records stored in SQLite	Do not assume automatic deletion solely from the retention setting; use backups and manual review.

20. Configure the AI inbox and Meta messaging webhook

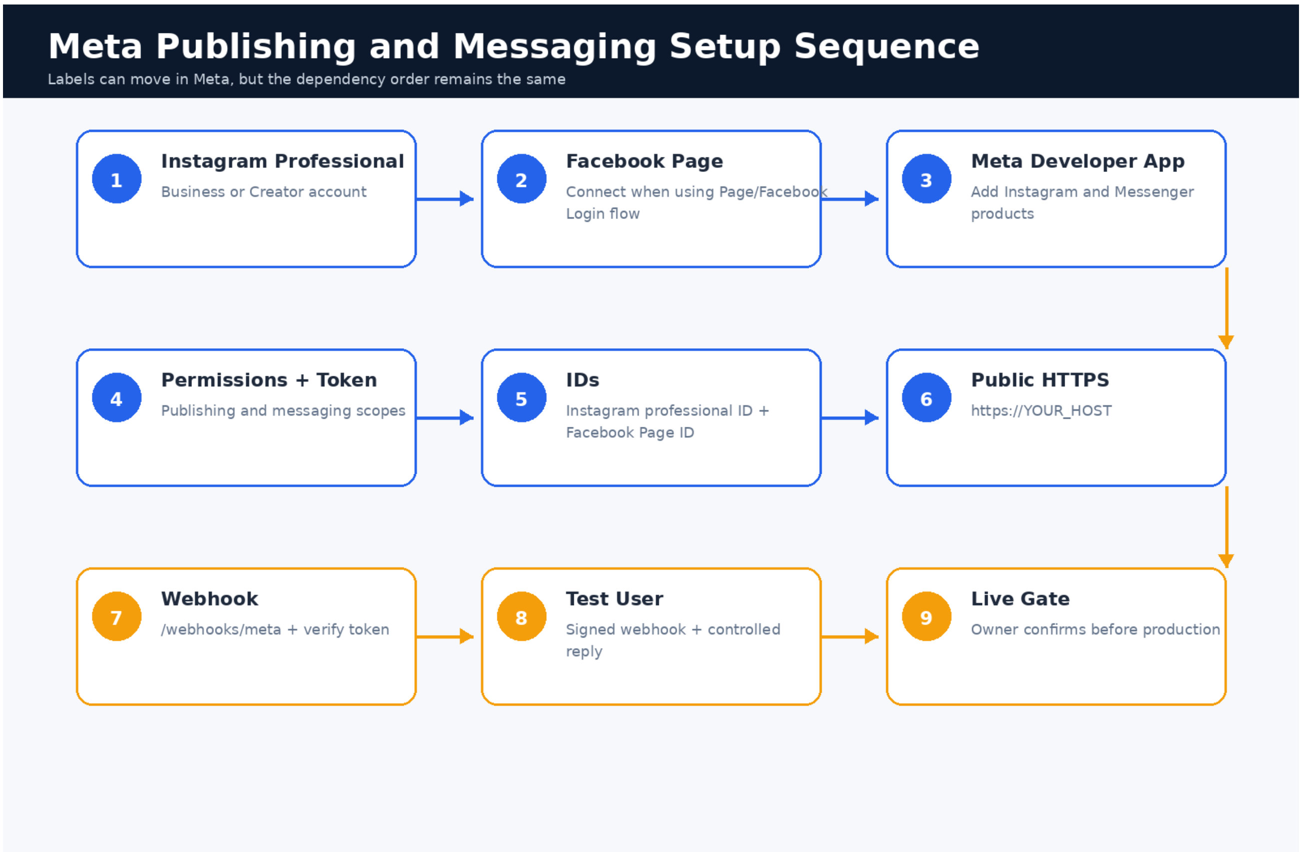


Figure 20-1. Correct order for Meta publishing and messaging setup.

20.1 Prepare the social accounts

1. Convert the Instagram account to a Professional account: Business or Creator.
2. For the Page/Facebook Login publishing flow, connect the Instagram professional account to the Facebook Page.
3. Create or use a Meta developer app that you control.
4. Add the Instagram and Messenger products required by the app flow.
5. Add your own accounts as app roles/testers while the app remains in development mode.

> **Account limitation**
> Instagram APIs do not manage ordinary consumer accounts. Messaging conversations begin only after the Instagram user messages the professional account. Group messaging is not supported.

20.2 Required permissions and identifiers

Purpose	Typical requirement	Value used by this app
Instagram content publishing	Instagram professional publishing permission for your selected login flow	INSTAGRAM_USER_ID + META_ACCESS_TOKEN
Facebook Page publishing	Page listing/read engagement/manage post permissions as required by Meta	FACEBOOK_PAGE_ID + META_ACCESS_TOKEN

Instagram messaging	instagram_business_basic and instagram_business_manage_messages for Instagram Login, or equivalent approved messaging permissions for your flow	META_ACCESS_TOKEN
Webhook authentication	Meta App Secret	META_APP_SECRET
Webhook challenge	Random value chosen by you and entered in both Meta and Oracle	META_WEBHOOK_VERIFY_TOKEN

20.3 Generate the webhook verification token

ORACLE TERMINAL

```
cd /opt/cryptopulse
openssl rand -hex 32
```

- Copy the complete random value.
- Enter the same value in the Meta webhook configuration and in Oracle .env.
- This is not the Meta App Secret. They are two different values.

20.4 Configure the Oracle .env safely

ORACLE TERMINAL

```
cd /opt/cryptopulse
cp .env .env.backup.$(date -u +%Y%m%dT%H%M%SZ)
nano .env
```

.ENV - FIRST MESSAGING CONFIGURATION

```
ENABLE_INSTAGRAM_MESSAGES=true
ENABLE_FACEBOOK_MESSAGES=false
AUTO_REPLY_ENABLED=false
META_APP_ID=YOUR_META_APP_ID
META_APP_SECRET=YOUR_META_APP_SECRET
META_WEBHOOK_VERIFY_TOKEN=YOUR_RANDOM_VERIFY_TOKEN
META_ACCESS_TOKEN=YOUR_LONG_LIVED_ACCESS_TOKEN
PROMOTION_DETECTION_ENABLED=true
PROMOTION_AUTO_REJECT_HIGH_RISK=true
PROMOTION_FINAL_ACCEPTANCE_REQUIRES_OWNER=true
```

Do not enable auto-replies yet
AUTO_REPLY_ENABLED=false still allows signed messages to be accepted, classified and recorded without sending a real reply. This is the safest first test.

20.5 Apply the configuration

ORACLE TERMINAL

```
cd /opt/cryptopulse
chmod 600 .env
docker compose config
docker compose up -d --build --remove-orphans
docker compose ps
docker compose exec app cryptopulse doctor
```

20.6 Add the webhook in Meta

6. Open the Meta developer dashboard for your app.
7. Open the Webhooks configuration for Instagram or the relevant messaging product.
8. Use callback URL: https://YOUR_PUBLIC_HOST/webhooks/meta
9. Paste the exact value stored in META_WEBHOOK_VERIFY_TOKEN.
10. Complete verification. Meta calls the GET endpoint and expects the challenge value back.
11. Subscribe the professional account/Page to supported message events such as messages and messaging_postbacks for your selected API flow.

TEST FROM WINDOWS, ORACLE OR A BROWSER

```
curl "https://YOUR_PUBLIC_HOST/webhooks/meta?hub.mode=subscribe&hub.verify_token=YOUR_TOKEN&hub.challenge=12345"
```

> **Expected result**
> The response body must be exactly 12345. HTTP 403 means the host, route or verify token is wrong.

20.7 Verify external reachability

ORACLE OR WINDOWS

```
curl https://YOUR_PUBLIC_HOST/health
curl https://YOUR_PUBLIC_HOST/ready
```

- Open the same health URL on your phone using mobile data, not the Oracle network.
- The certificate must be trusted; do not continue with a browser certificate warning.
- Caddy requires ports 80 and 443 in both OCI ingress rules and the server firewall.

21. Test replies as real users would use the account

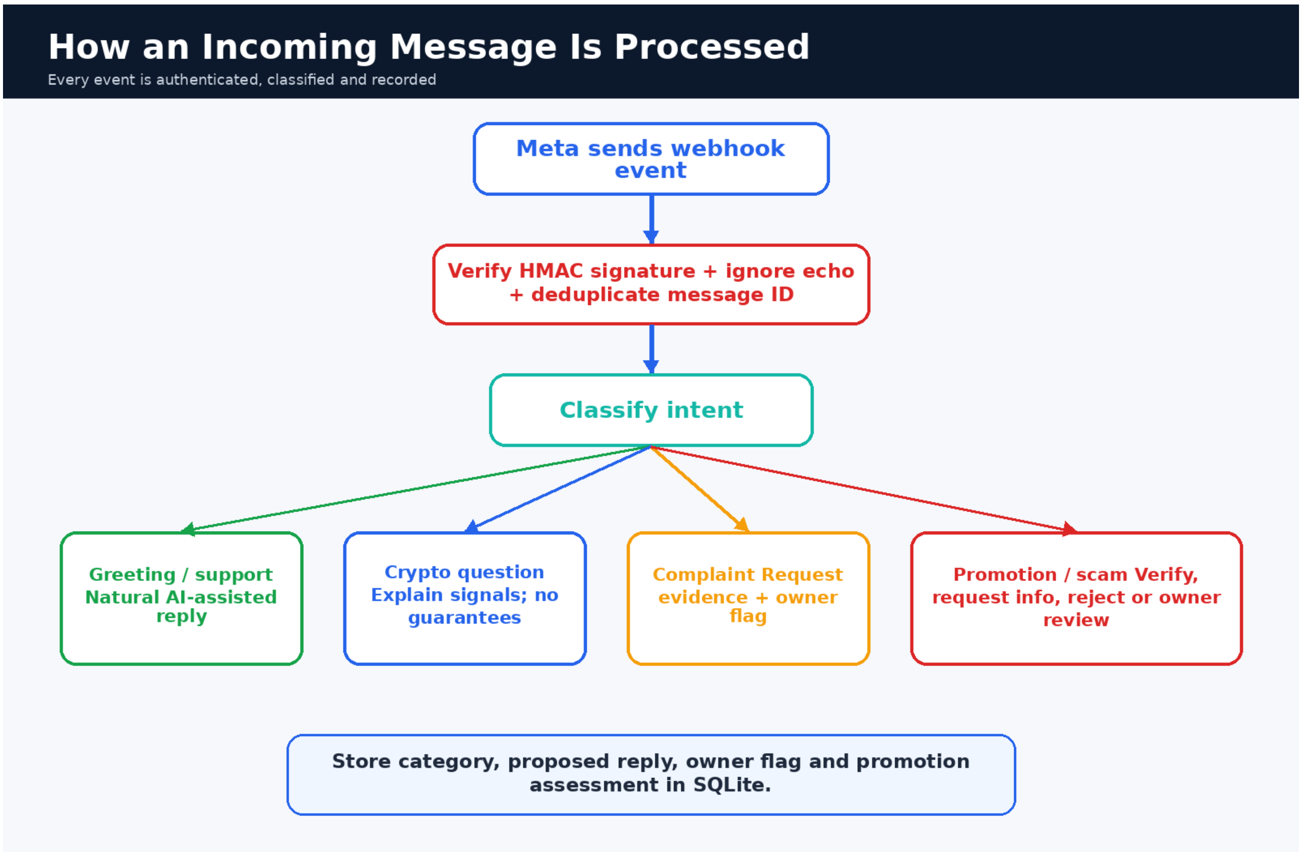


Figure 21-1. Incoming message authentication, classification and reply planning.

21.1 Test with automatic replies disabled

12. Keep DRY_RUN=true and AUTO_REPLY_ENABLED=false.
13. Send a direct message from a Meta test user to the professional account.
14. Watch the application logs.
15. Confirm the webhook receives one event and does not reject the signature.
16. Confirm the event is recorded in the conversations table.
17. Send the same webhook again or let Meta retry it; the duplicate message ID must not create a second row.

ORACLE TERMINAL

```
cd /opt/cryptopulse
docker compose logs -f app
```

21.2 Inspect recent conversations

ORACLE TERMINAL

```
docker compose exec app python -c "from cryptopulse.config import get_settings; from cryptopulse.db import
Database,ConversationRow; db=Database(get_settings()); s=db.session();
rows=s.query(ConversationRow).order_by(ConversationRow.created_at.desc()).limit(20).all();
[print(r.created_at,r.platform,r.category,r.requires_owner,repr(r.inbound_text),repr(r.reply_text),r.promotion_jso
n) for r in rows]"
```

> **Privacy warning**
> This command prints message text and promotion details to your terminal. Do not paste that output into public chats, issues

> or screenshots.

21.3 Real-user test matrix

Test message	Expected classification	Expected behavior
Hello, good morning	greeting	Transparent AI-assisted greeting.
Will Bitcoin definitely rise?	crypto_question	No guarantee; asks which coin or post.
Your post is wrong	complaint	Requests post date/screenshot and marks owner review.
We want a paid collaboration	promotion_offer	Requests missing company, campaign and contract information.
Send your seed phrase / pay a verification fee	scam	Rejects and does not reveal secrets.
Random support question	general_support	Requests more detail.

21.4 Enable a controlled dry-run reply

ORACLE TERMINAL

```
nano /opt/cryptopulse/.env
```

.ENV

```
DRY_RUN=true
AUTO_REPLY_ENABLED=true
```

ORACLE TERMINAL

```
cd /opt/cryptopulse
docker compose up -d --force-recreate app
docker compose logs -f app
```

> **Dry-run behavior**
> The app produces a simulated message ID rather than contacting the real Meta Send API. This validates routing and database persistence safely.

21.5 First live auto-reply canary

18. Use only an app-role/test account.
19. Confirm the Meta access token and messaging permissions are valid.
20. Keep live social publishing switches false while testing messaging.
21. Set DRY_RUN=false and AUTO_REPLY_ENABLED=true.
22. Send one greeting from the test user.
23. Confirm exactly one reply appears and the database contains one record.
24. Immediately return to AUTO_REPLY_ENABLED=false if a duplicate or incorrect reply occurs.

EMERGENCY STOP FOR REPLIES

```
sed -i 's/^AUTO_REPLY_ENABLED=true/AUTO_REPLY_ENABLED=false/' /opt/cryptopulse/.env
cd /opt/cryptopulse && docker compose up -d --force-recreate app
```

22. Promotion verification and owner review

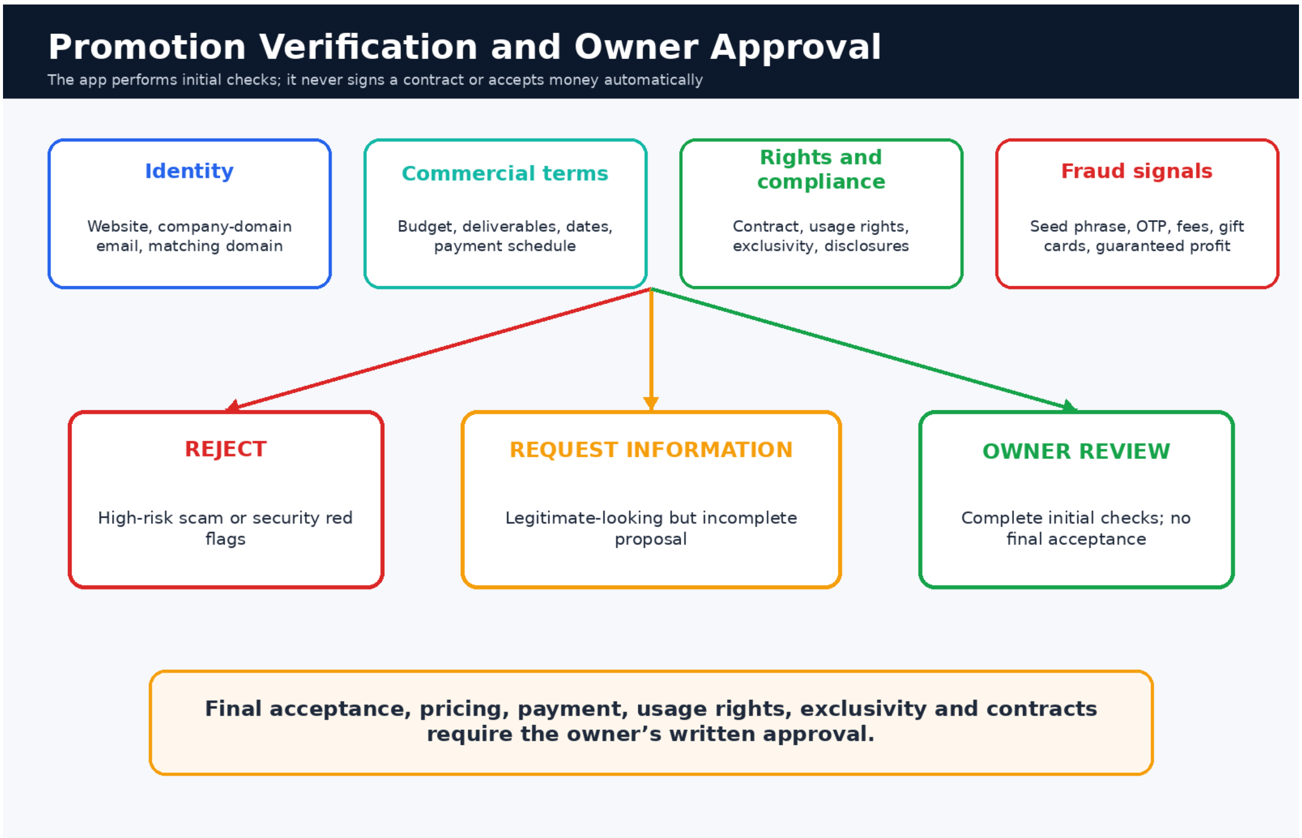


Figure 22-1. Promotion screening decisions and mandatory owner approval.

22.1 What the application checks

- Website URL present in the message.
- Contact email present and whether its domain appears to match the website domain.
- Budget amount or currency expression.
- Requested deliverables such as Reel, Story, post or carousel.
- Contract, usage-rights or license language.
- Campaign date or deadline language.
- Red flags such as private key, seed phrase, OTP, processing fee, gift card, guaranteed profit, buy first or urgent payment.

> **Initial screening only**
> The current release does not browse company registries, run malware scanning, verify beneficial ownership, validate a token contract on-chain, or negotiate legal terms. Treat its score as triage, not proof that an advertiser is legitimate.

22.2 Possible outcomes

Outcome	When it happens	System reply / owner action
Reject	High-risk red flags or high scam score	Declines without revealing fraud-detection rules.
Request information	Website, domain email, budget, deliverables, contract/rights or dates missing	Asks advertiser to provide the missing campaign information.
Owner review	Initial fields are present and no high-risk	States that no campaign is accepted until

	indicator is found	written owner approval.

22.3 Review owner-required records

ORACLE TERMINAL

```
docker compose exec app python -c "from cryptopulse.config import get_settings; from cryptopulse.db import
Database,ConversationRow; db=Database(get_settings()); s=db.session();
rows=s.query(ConversationRow).filter(ConversationRow.requires_owner.is_(True)).order_by(ConversationRow.created_at
.desc()).limit(30).all(); [print(r.id,r.created_at,r.category,repr(r.inbound_text),r.promotion_json) for r in
rows]"
```

22.4 Owner due-diligence checklist

- Contact the company through a channel published on its official website, not only through the DM.
- Confirm the sender controls the company-domain email.
- Inspect the contract, payment schedule, cancellation terms, usage rights and exclusivity.
- For crypto promotions, independently verify the official domain, contract address, audits, liquidity and regulatory/advertising requirements.
- Reject any request for your password, OTP, API key, Meta token, private key, seed phrase, remote computer access or advance payment.
- Use clear sponsored-content disclosure and Meta paid-partnership tools when applicable.

> **Non-negotiable rule**
> Final advertising acceptance, pricing, payment, contract, exclusivity and content rights remain manual owner decisions.

23. Operate, inspect and recover the inbox subsystem

23.1 Daily health commands

ORACLE TERMINAL

```
cd /opt/cryptopulse
docker compose ps
curl https://YOUR_PUBLIC_HOST/health
curl https://YOUR_PUBLIC_HOST/ready
docker compose logs --since=24h app | tail -n 400
```

23.2 What to monitor

- Webhook HTTP 403 errors - usually wrong signatures, app secret or non-Meta requests.
- Repeated sender/message IDs - the database should deduplicate by message_id.
- Meta 400/401/403 responses - token, permission, recipient or messaging-window issue.
- Unexpected categories or reply text - disable auto-replies and inspect the stored record.
- Database size and disk usage.
- Token expiry and Meta app review status.

23.3 Back up before configuration changes

ORACLE TERMINAL

```
cd /opt/cryptopulse
bash scripts/backup_db.sh
cp .env .env.backup.$(date -u +%Y%m%dT%H%M%SZ)
```

23.4 Emergency safe mode

ORACLE TERMINAL

```
cd /opt/cryptopulse
sed -i 's/^DRY_RUN=false/DRY_RUN=true/' .env
sed -i 's/^AUTO_REPLY_ENABLED=true/AUTO_REPLY_ENABLED=false/' .env
sed -i 's/^PUBLISH_INSTAGRAM=true/PUBLISH_INSTAGRAM=false/' .env
sed -i 's/^PUBLISH_FACEBOOK=true/PUBLISH_FACEBOOK=false/' .env
docker compose up -d --force-recreate app
docker compose exec app cryptopulse doctor
```

23.5 Common inbox errors

Problem	Likely cause	Fix
Webhook verification returns 403	Verify token differs between Meta and .env	Copy the exact same token, recreate app container, retry challenge.
POST webhook returns 403	Wrong META_APP_SECRET or missing x-hub-signature-256	Confirm the app secret and test only with Meta-signed events.
Event accepted but no reply	AUTO_REPLY_ENABLED=false, dry-run, unsupported/echo event, or sender missing	Check .env, logs, message payload and conversation row.
Meta Send API rejects reply	Expired token, missing messaging permission, wrong API flow/host, or conversation not eligible	Regenerate token, verify scopes and test with an app-role user.

Duplicate replies	Multiple live instances or message IDs not stable	Disable replies, confirm one deployment, inspect message_id values.
Promotion marked legitimate too easily	Text-only initial screening	Require owner due diligence; do not auto-accept.

23.6 Key rotation

25. Disable auto-replies and live publishing.
26. Create or rotate the token/secret in the provider dashboard.
27. Back up .env and database.
28. Replace only the required value in /opt/cryptopulse/.env.
29. Recreate the app container and run cryptopulse doctor.
30. Run a dry-run or test-user canary before restoring live operation.

24. Final production acceptance checklist

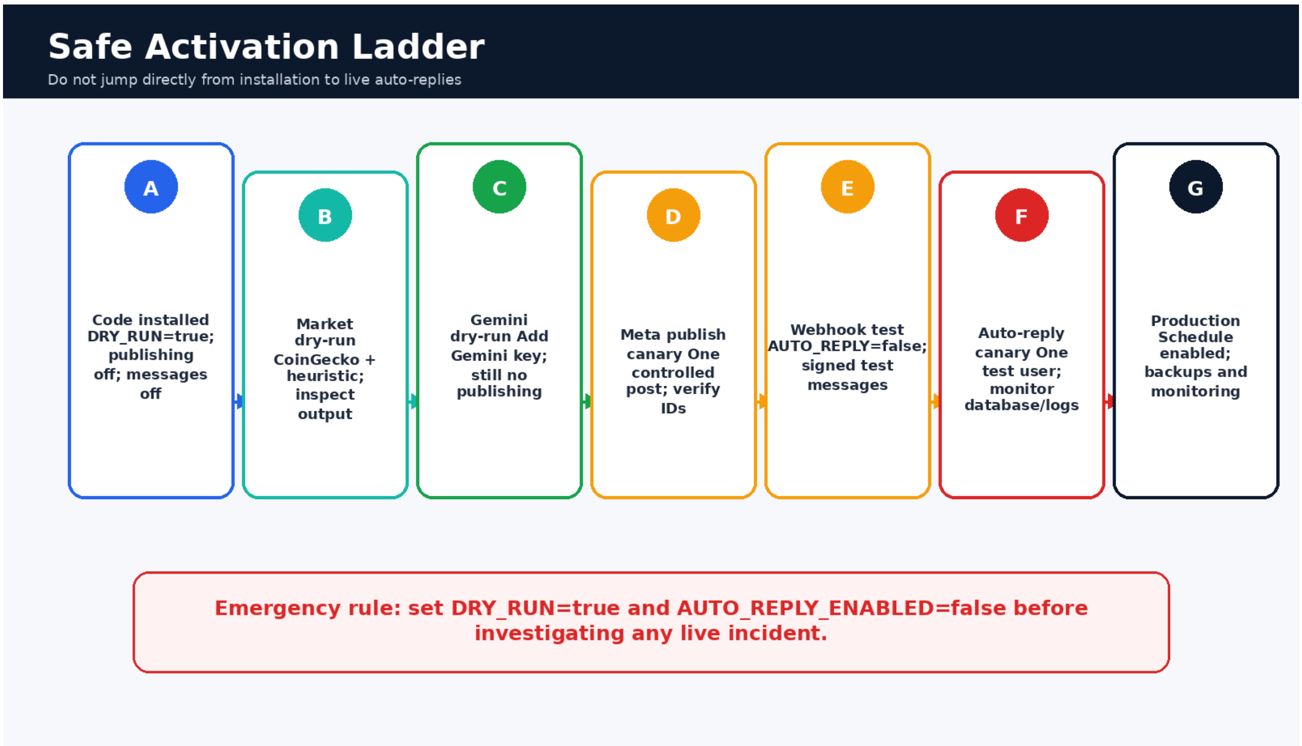


Figure 24-1. Required activation order from installation to production.

24.1 Infrastructure and code

- GitHub main branch contains the flat root structure and CI is green.
- Oracle app and Caddy containers are healthy.
- Ports 22, 80 and 443 are open; port 8080 is bound only to localhost.
- HTTPS works from a phone on mobile data.
- Database backup and a known-good Git commit are recorded.

24.2 Market and publishing

- CoinGecko dry-run completes without insufficient-asset or rate-limit failure.
- Gemini dry-run produces acceptable analysis and captions.
- Generated bullish and bearish images are manually inspected.
- A signed media URL is reachable externally.
- One Instagram/Facebook canary post returns valid post IDs and appears correctly.

24.3 Messaging and promotions

- Meta webhook GET verification returns the exact challenge.
- Invalid webhook signatures receive HTTP 403.
- A test-user message creates one database row.
- Echo events and duplicate message IDs do not create duplicate replies.
- Greeting, crypto question, complaint, incomplete promotion, complete promotion and scam scenarios behave as expected.
- Final promotion acceptance is still owner-approved.
- AUTO_REPLY_ENABLED is enabled only after a controlled live canary.

> **Production approval**
> Do not enable all publishing and messaging switches at the same time. Change one integration, test it, observe it, and only

then enable the next.

Appendix C - Full v0.2.0 configuration profiles

C.1 Safe first deployment

.ENV PROFILE

```
APP_ENV=production
LOG_LEVEL=INFO
DRY_RUN=true
TIMEZONE=UTC
SCHEDULE_HOURS=0,6,12,18
RUN_ON_STARTUP=false
MAX_CONCURRENT_WORKERS=4
MAX_MARKET_PAGES=80
MARKET_PAGE_SIZE=250
MIN_MARKET_CAP_USD=1000000
MIN_VOLUME_24H_USD=100000
TOP_PRELIMINARY_PER_SIDE=15
TOP_FINAL_PER_SIDE=5
PREDICTION_HORIZON_HOURS=6

MARKET_PROVIDER=coingecko
COINGECKO_API_KEY=
COINGECKO_BASE_URL=https://api.coingecko.com/api/v3
COINGECKO_REQUEST_INTERVAL_SECONDS=2.2
NEWS_PROVIDER=hybrid

LLM_PROVIDER=heuristic
GEMINI_API_KEY=
GEMINI_MODEL=gemini-3.5-flash
OPENAI_API_KEY=
OPENAI_MODEL=gpt-5-mini
LLM_BATCH_SIZE=10
LLM_MAX_CONCURRENCY=2
LLM_TIMEOUT_SECONDS=90

META_GRAPH_VERSION=v25.0
META_ACCESS_TOKEN=
INSTAGRAM_USER_ID=
FACEBOOK_PAGE_ID=
PUBLISH_INSTAGRAM=false
PUBLISH_FACEBOOK=false
PUBLISH_STORIES=false
PUBLISH_REELS=false

PUBLIC_HOST=localhost
PUBLIC_BASE_URL=
MEDIA_SIGNING_SECRET=change-this-to-a-long-random-secret
MEDIA_URL_TTL_SECONDS=86400
MEDIA_DELETE_GRACE_HOURS=24

DATABASE_URL=sqlite:///./data/cryptopulse.db
MEDIA_ROOT=./data/media
RETAIN_RAW_DATA=false
RETAIN_GENERATED_MEDIA=true
PROJECT_PROFILE_REFRESH_HOURS=168

BRAND_NAME=CryptoPulse AI
BRAND_HANDLE=@yourhandle
DISCLAIMER=Educational market intelligence only. Not financial advice. Crypto assets are highly volatile.

ENABLE_INSTAGRAM_MESSAGES=false
ENABLE_FACEBOOK_MESSAGES=false
AUTO_REPLY_ENABLED=false
META_APP_ID=
META_APP_SECRET=
META_WEBHOOK_VERIFY_TOKEN=
CONVERSATION_RETENTION_DAYS=30
PROMOTION_DETECTION_ENABLED=true
PROMOTION_AUTO_REJECT_HIGH_RISK=true
PROMOTION_FINAL_ACCEPTANCE_REQUIRES_OWNER=true
PROMOTION_MIN_IDENTITY_SCORE=75
PROMOTION_MIN_BRAND_SAFETY_SCORE=80
```

C.2 Controlled live messaging canary

.ENV - USE ONLY AFTER TEST WEBHOOKS PASS

```
DRY_RUN=false

# Keep publishing disabled while testing messages
PUBLISH_INSTAGRAM=false
PUBLISH_FACEBOOK=false
PUBLISH_STORIES=false
PUBLISH_REELS=false

ENABLE_INSTAGRAM_MESSAGES=true
ENABLE_FACEBOOK_MESSAGES=false
AUTO_REPLY_ENABLED=true

META_APP_ID=YOUR_APP_ID
META_APP_SECRET=YOUR_APP_SECRET
META_WEBHOOK_VERIFY_TOKEN=YOUR_RANDOM_VERIFY_TOKEN
META_ACCESS_TOKEN=YOUR_LONG_LIVED_TOKEN

PUBLIC_HOST=YOUR_PUBLIC_HOST
PUBLIC_BASE_URL=https://YOUR_PUBLIC_HOST
MEDIA_SIGNING_SECRET=YOUR_LONG_RANDOM_MEDIA_SECRET

PROMOTION_DETECTION_ENABLED=true
PROMOTION_AUTO_REJECT_HIGH_RISK=true
PROMOTION_FINAL_ACCEPTANCE_REQUIRES_OWNER=true
```

Why publishing remains disabled
Messaging and social publishing are separate risks. Validate the inbox first, then restore publishing in a later controlled change.

C.3 Normal operational commands

COMMAND QUICK REFERENCE

```
cd /opt/cryptopulse

# Status and health
docker compose ps
curl https://YOUR_PUBLIC_HOST/health
curl https://YOUR_PUBLIC_HOST/ready

# Logs
docker compose logs -f app
docker compose logs --since=1h app

# One market/content cycle
docker compose exec app cryptopulse run-once

# Validate configuration
docker compose exec app cryptopulse doctor

# Update code
bash scripts/update_from_git.sh main

# Database backup
bash scripts/backup_db.sh

# Restart app only
docker compose restart app

# Stop everything
docker compose down
```

Official sources for v0.2.0

Provider interfaces change. The application defaults and this manual were reviewed on 20 July 2026. Use official documentation when permissions, model names or dashboard labels differ.

[S18] Google - Gemini API Getting Started: https://ai.google.dev/gemini-api/docs/get-started

[S19] Google - Using Gemini API Keys: https://ai.google.dev/gemini-api/docs/api-key

[S20] CoinGecko - Setting Up Your API Key: https://docs.coingecko.com/docs/setting-up-your-api-key

[S21] CoinGecko - Keyless Public API: https://docs.coingecko.com/docs/keyless-public-api

[S22] Oracle - Always Free Resources: https://docs.oracle.com/en-us/iaas/Content/FreeTier/freetier_topic-Always_Free_Resources.htm

[S23] Meta official Instagram API workspace - overview and requirements: https://www.postman.com/meta/instagram/documentation/23987686-9386f468-7714-490f-9bfc-9442db5c8f00

[S24] Meta official Instagram Send API: https://www.postman.com/meta/instagram/folder/23987686-f05b6c9f-a4be-4511-9f88-1cd94828fdf3

[S25] Meta official Instagram Messaging Webhook: https://www.postman.com/meta/instagram/request/23987686-95cce6f6-b811-41dc-b560-d43741c5002a

[S26] Meta official Instagram Conversations API: https://www.postman.com/meta/instagram/folder/23987686-6a91368f-1fa8-4614-9ed6-7d1e08c21e62

[S27] OpenAI - ChatGPT Plus and API usage are separate: https://help.openai.com/en/articles/6950777-what-is-chatgpt-plus